Week 5 Sedimentary Structures

Lecture overview:

- Classification of structures
- Physical primary structures
- Stratification
  - What is a bed?
  - Identification of bedding surfaces
- Origin of stratification
- Stratification of unidirectional flows
  - Cross stratification
  - Planar stratification
  - Cross stratification produced by waves and wind
- Other 'intrabed' structures/textures
  - Graded bedding
  - Massive beds
  - Imbrication
- Sole marks
  - Flute casts
  - Tool marks
- Secondary structures
- Biogenic structures



Emphasis today on recognition and then interpretation

Why do we care?



Observing and correctly identifying individual sedimentary structures and bedforms allow us to interpret the palaeohydraulic conditions under which they formed

Studying vertical (and lateral) sequences, allow us to interpret changes in space and time in:

- direction of flow ('palaeotransport direction')
- strength of flow and flow depth
- duration and variability of flow

And therefore to interpret:

- depositional environments
- changing climate and sea levels
- provenance and tectonic processes

Classification of Sedimentary Structures

Definition: Features, on the scale of hand specimens to large outcrops, produced within a depositional environment, during or (usually) not long after deposition

Classified on basis of:

- Kind of mechanism that produces them (i.e., physical, chemical, or biological)
- Time of development relative to time of deposition (i.e., primary or secondary)



	physical	chemical	biological
primary	stratification sole marks	stratification	biotubation (tracks, trails)
secondary	deformation intrusion desiccation	nodules concretions deformation stylolites	bioturbation (burrows)









Physical Structures

- Form during erosion, transport, deposition or early post-depositional processes
  - Primary
    - Stratification formed in cohensionless sediment (syn-depositional, intrabed structures) by migration of bedforms and settling of sediment
    - Sole marks formed in cohesive sediment (syn-depositional, interbed structures) by erosion and/or transportation
  - Secondary
    - Post-depositional structures produced by 'soft-sediment' deformation, intrusion and desiccation
- Found within beds and on bedding planes



Syn-depositional intrabed structures

Syn-depositional interbed structures

Stratification



- Stratification: layering brought about by deposition

- Caused by changes in depositional conditions with time

- Layering, aka stratification, aka bedding, is not exclusive to sedimentary rocks. Also occurs in lavas, plutonic and metamorphic rocks

- Two key questions to consider:
  - What it was about depositional conditions that changed with time to give rise to stratification?
  - What it is about the rock itself that makes the stratification manifest?

- To identify stratification look for vertical changes from bed to bed in:
  - grain size (texture)
  - composition (maybe indicated by colour, differential weathering)
  - structure
  - or, more rarely, grain shape, packing, or orientation

Terminology

Stratification is officially divided into bedding and lamination based on the thickness of the strata

BEDDING (bed, beds)	very thick-bedded	100 cm
	thick-bedded	30 cm
	medium-bedded	10 cm
	thin-bedded	3 cm
	very thin-bedded	1 cm
LAMINATION (lamina, laminae)	laminated	0.3 cm
	thinly laminated	

Definition of beds based on thickness (defined for sandstones)

Beds and bedding surfaces

Beds are bounded above and below by bedding surfaces or planes
- Can be readily recognised where defined by changes in lithology

Bedding surfaces are produced during periods of non-deposition, or abrupt change in depositional conditions. Non-deposition commonly associated with erosion

Surfaces are essentially time synchronous (form in hours, days, years = instant in geologic time); therefore, a bed is a time-stratigraphic unit of limited areal extent and relatively short time span

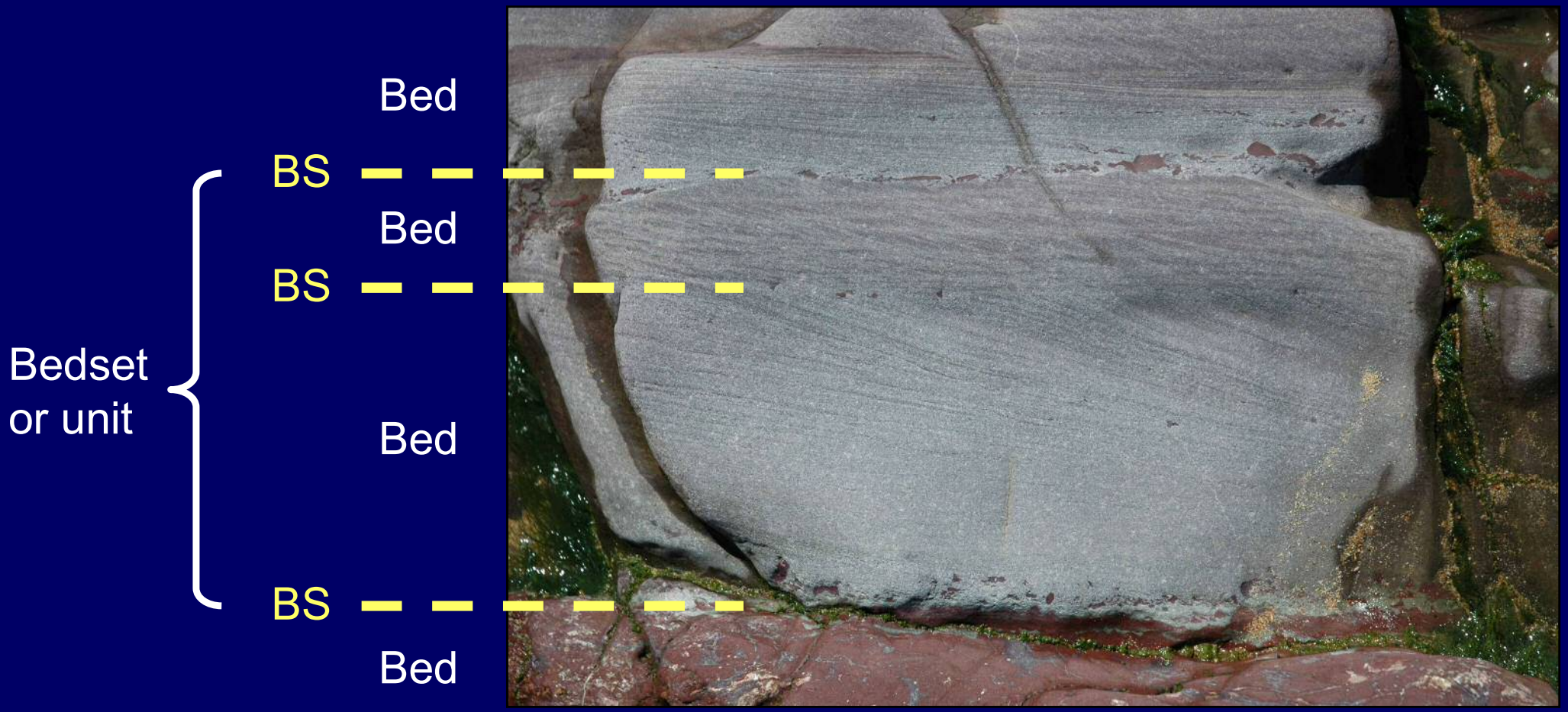


Recognition of bedding surfaces and beds

Distinction of beds depends on recognition of bedding surfaces

Recognition of bedding surfaces based on:

- change in lithology
- abrupt change in sedimentary structures
- rip-up clast horizons



Recognition of bedding surfaces and beds

Recognition of bedding surfaces based on:

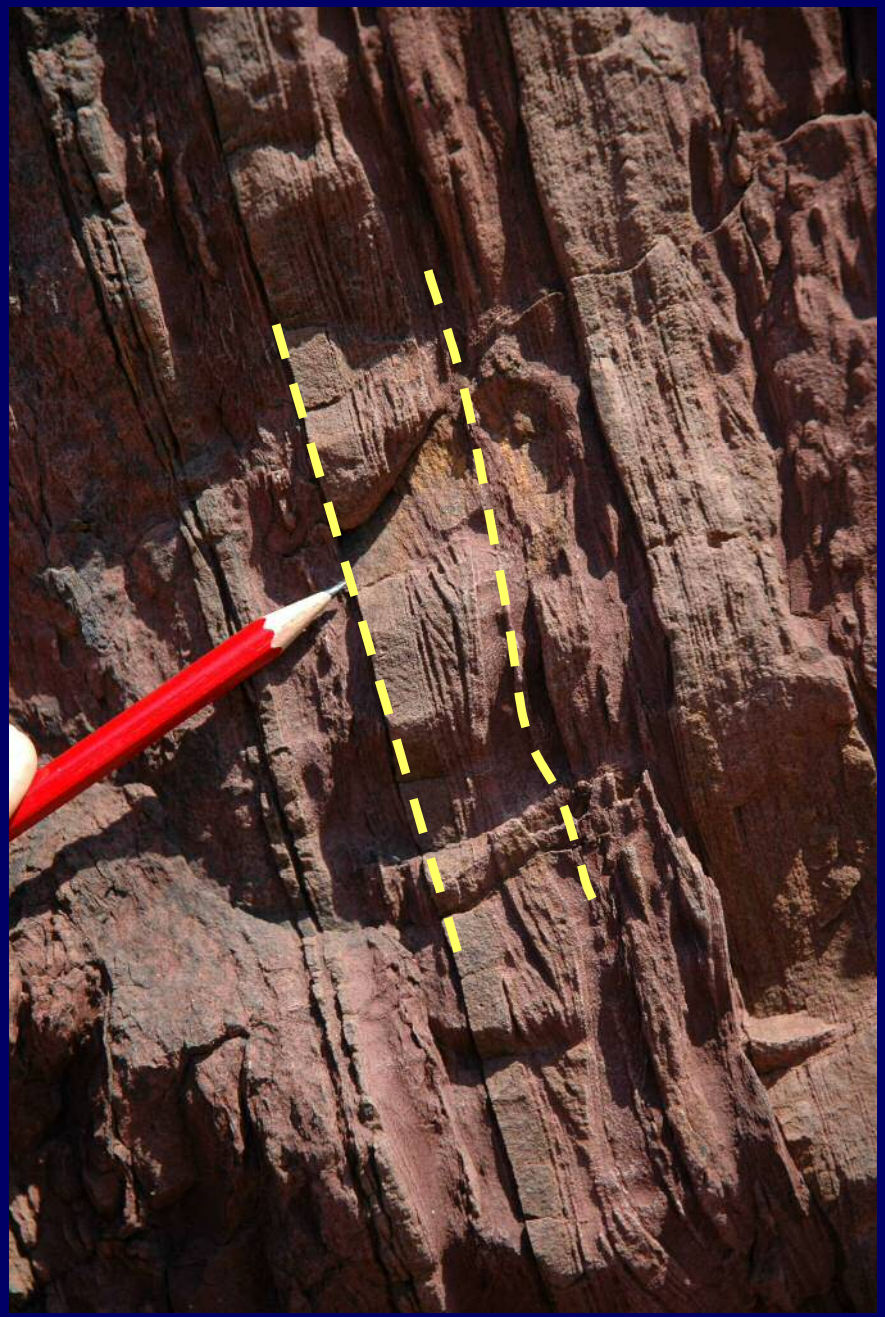
- trails and trackways of organisms





Thalassinoides

Process based definition of what is a bed



Another way to think about a bed: depositional product of one flow event = bed

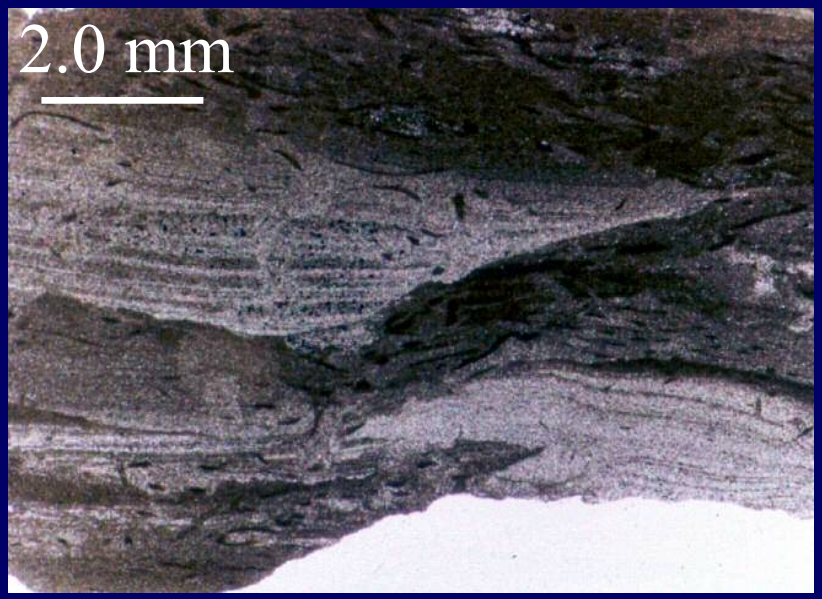
Origin of Stratification

Two major “scenarios” for the origin of stratification we will consider

1. Quiet fluid deposition of particles by settling



- Ocean bottom (plus lakes) mainly (but also rivers etc)
- Low velocity currents carry sediments from upcurrent
- Usually fine grained (but not always)
- Usually thin lamination
- Usually nearly or perfectly even or planar
- Often later bioturbated (can obliterate original laminations)



Bioturbation in mudstones disrupts the original depositional textures

Origin of Stratification

2. Deposition of particles by tractional currents

- Deposition during bed-load (or bed-load plus suspended load) transport
- Moderate to strong currents
- Stratification thick to thin depending on nature of variations in sediment supply, currents and deposition rate
- Even (parallel) lamination and cross stratification
- Usually coarsest silt to gravel range



Even (parallel) lamination

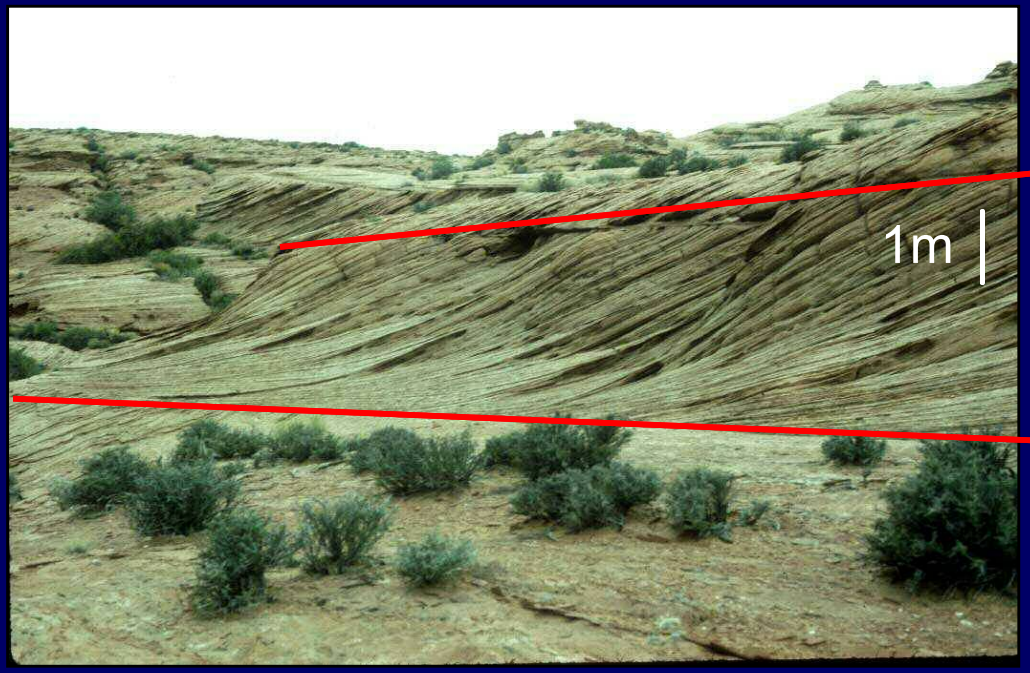


Cross stratification

Cross Stratification

Definition: all stratification that is locally at some angle to the overall stratification due to primary processes (e.g. excluding tectonic tilting)





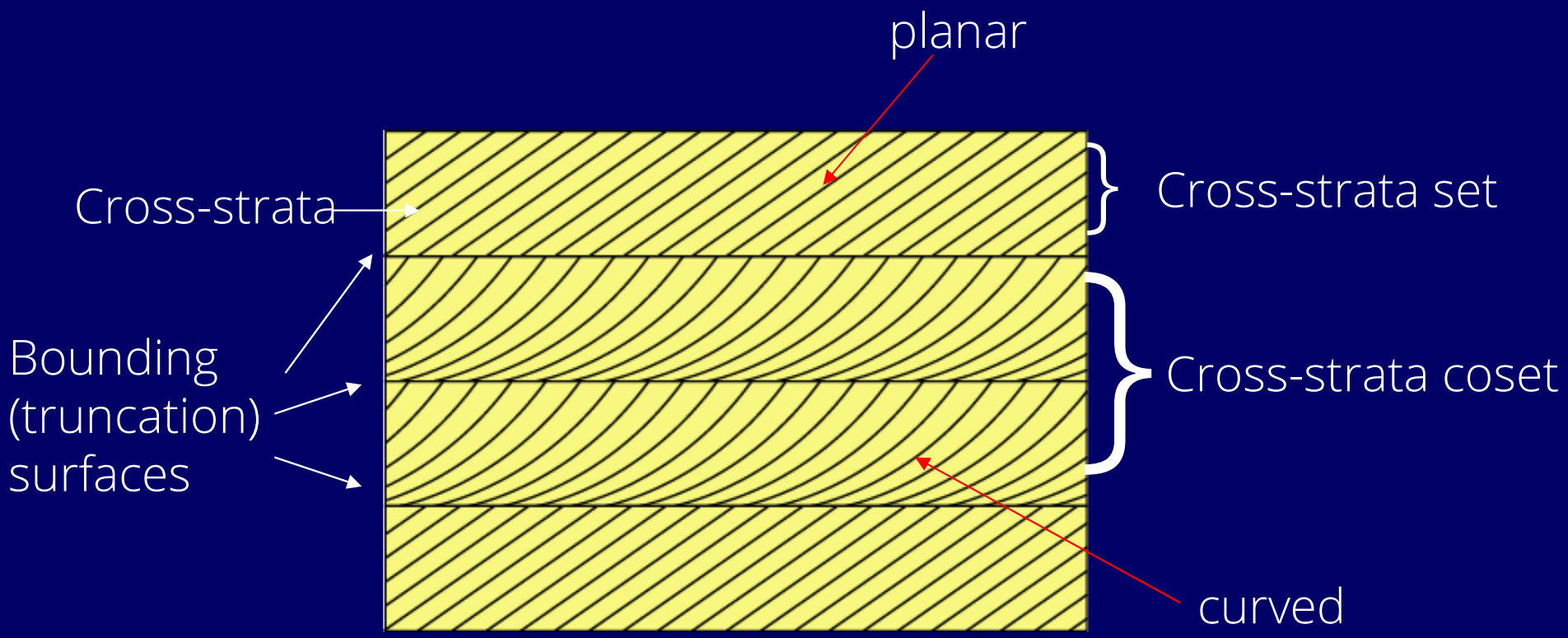
- Vertical scale varies from mm to several metres
- Geometry varies enormously
- Interpretation of cross stratification is well advanced ➡ probably the single most useful tool in interpreting the physical aspects of loose-sediment depositional environments

Geometrical Terminology

Cross-strata: inclined strata normally between bounding surfaces which are erosional surfaces that truncate underling cross-strata when present.

Cross-strata set: a single set of similar cross-strata

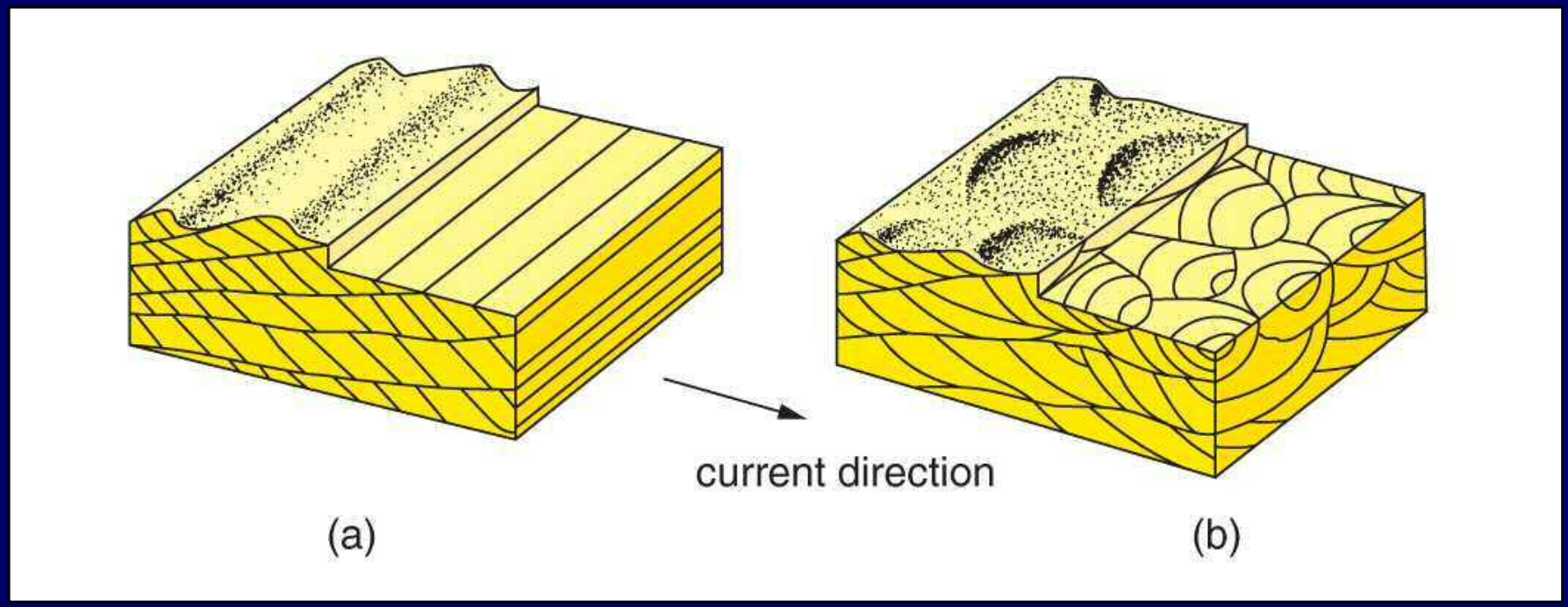
Cross-strata co-set: sets of cross-strata of the same type in vertical sequence and separated by bounding surfaces.



Cross-stratification

Orientation matters

The geometry of cross stratification commonly looks different in differently oriented sections normal to the overall plane of stratification



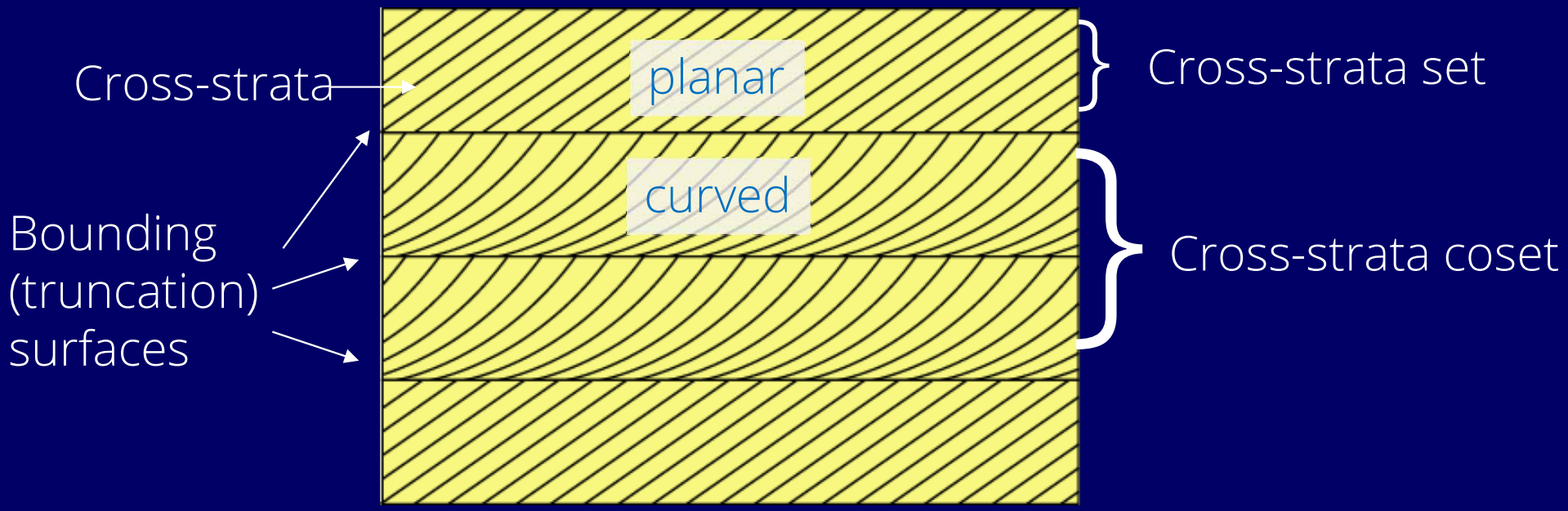
View the cross stratification on as many differently oriented planes normal to bedding as you can, because it might look quite different depending on the direction

Geometrical Terminology

Cross-strata: inclined strata normally between bounding surfaces which are erosional surfaces that truncate underling cross-strata when present.

Cross-bed: strata produced by the migration of a dune

Cross-lamination: strata produced by the migration of a ripple



Cross-stratification

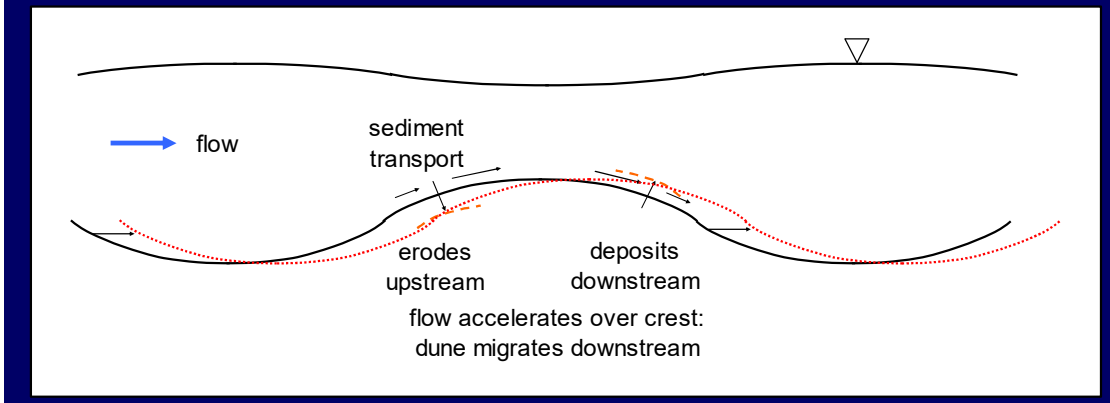
Practice Example



Label the image to show the following:
truncation surfaces, set, coset

How bed forms make cross stratification

- Water or air moving over loose (cohesionless) sediment moulds the bed into 'bedforms' of varying topography
- Bedforms can be anything from mounds a few cm across and less than a cm high, to large structures more than 10m high



Water surface is out of phase with the bed

➢Depth variation is out of phase with the bed

•Flow accelerates from trough to crest

•Sediment transport increases from trough to crest

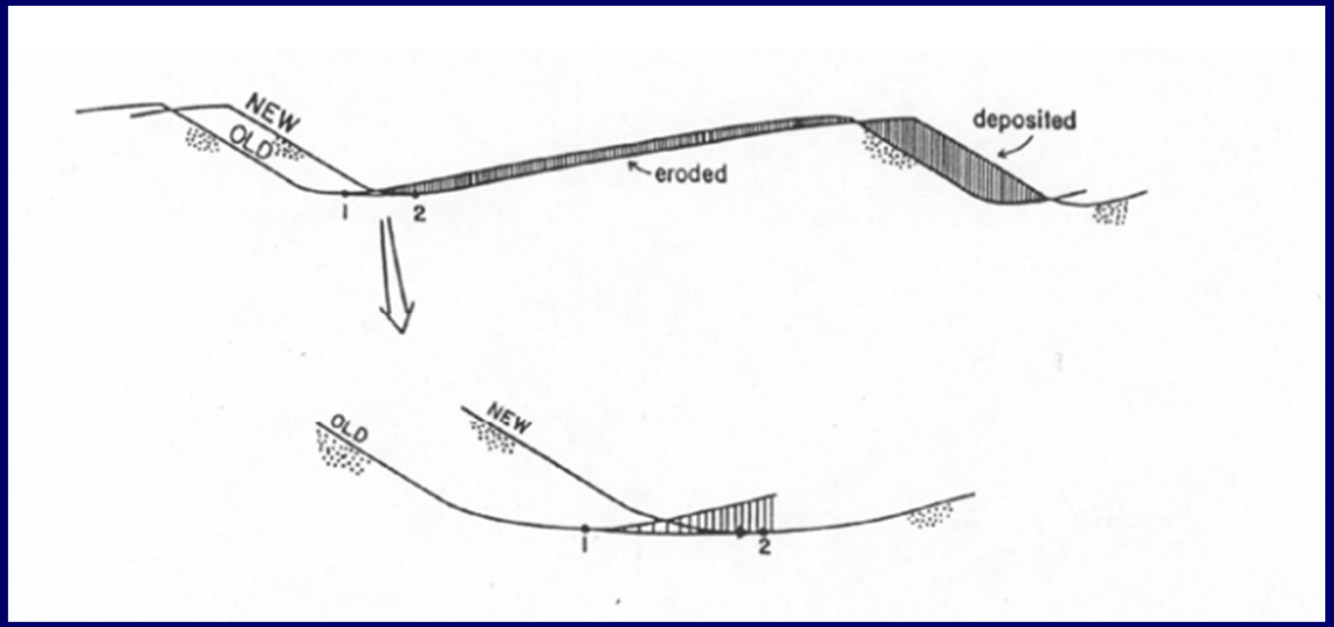
•Bedform migrates downstream

Ripples migrate downstream as a train of ripples



How bed forms make cross stratification

Cross stratification is formed by the erosion and deposition associated with a train of bed forms



As the trough of the upstream ripple moves downstream, material is eroded from the stoss and deposited on the lee face of the downstream ripple

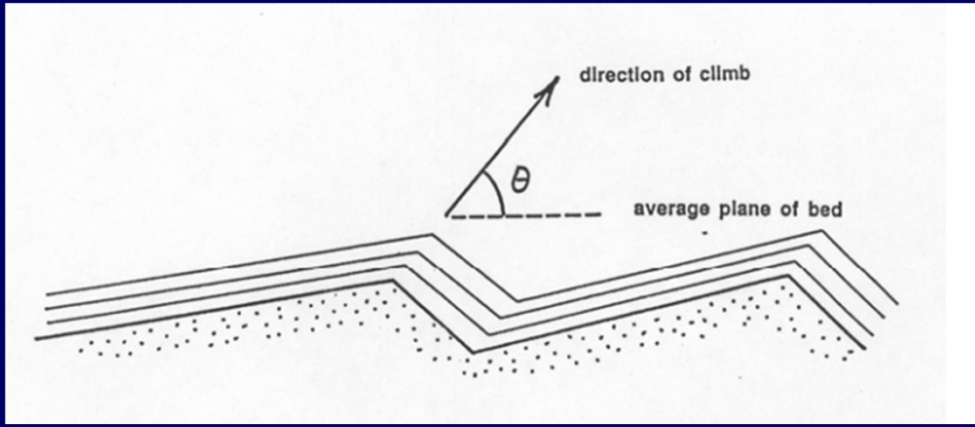
If no new material is added to the bed while the ripples move, then the average elevation of the bed does not change with time

- the foresets deposited by a downstream ripple are entirely eroded away again as the next trough upstream passes by

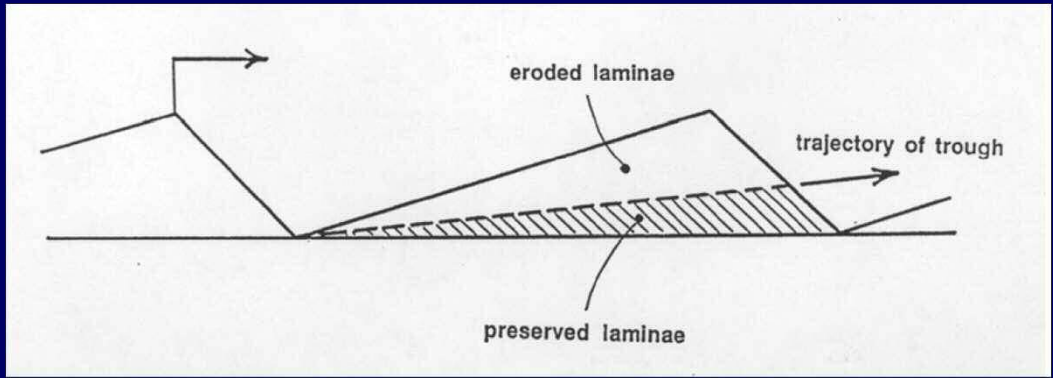
If bedform stopped moving abruptly, you'd see most recent position of the lee faces preserved

Origin of angle of climb

If new sediment is added everywhere to build the bed upward, the ripples will have a component of upward movement



The angle of climb (θ) of the ripples depends on the ratio of rate of bed aggradation to speed of ripple movement



As the ripples climb in space, their troughs climb with them.

The erosion surface associated with the downstream movement of a given trough passes above the erosion surface that was formed when the preceding trough passed by.

The lowest parts of the foresets deposited by the ripple that was located between those two troughs are then preserved rather than eroded entirely .

This remnant set is bounded both above and below by erosion surfaces.

Climbing cross stratification



Climbing ripple cross stratification

At high angles of climb, the entire ripple profile is preserved, and there are no erosion surfaces in the deposit

At low angles of climb, only the lower parts of foreset deposits are preserved

- the individual sets are bounded by erosion surfaces

The height of the sets is always less than the height of the bed forms that were responsible for the cross stratification

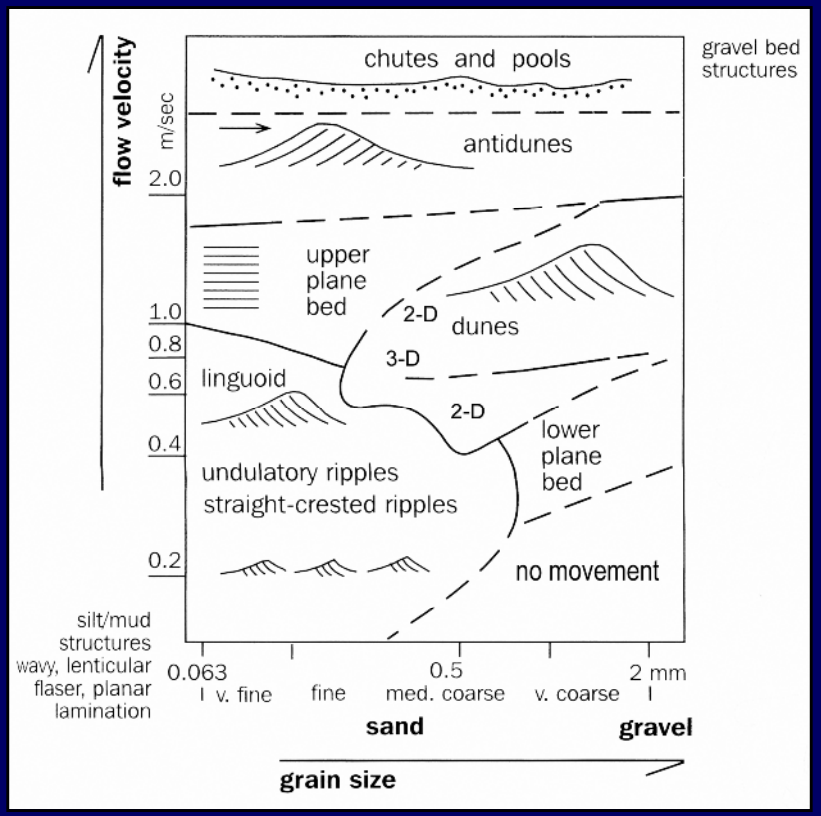
Important kinds of cross stratification

(i) Unidirectional flow cross stratification

- Small scale (formed by ripples)
- Large scale (formed by dunes)



Empirically determine the relationship between bedforms and current velocity

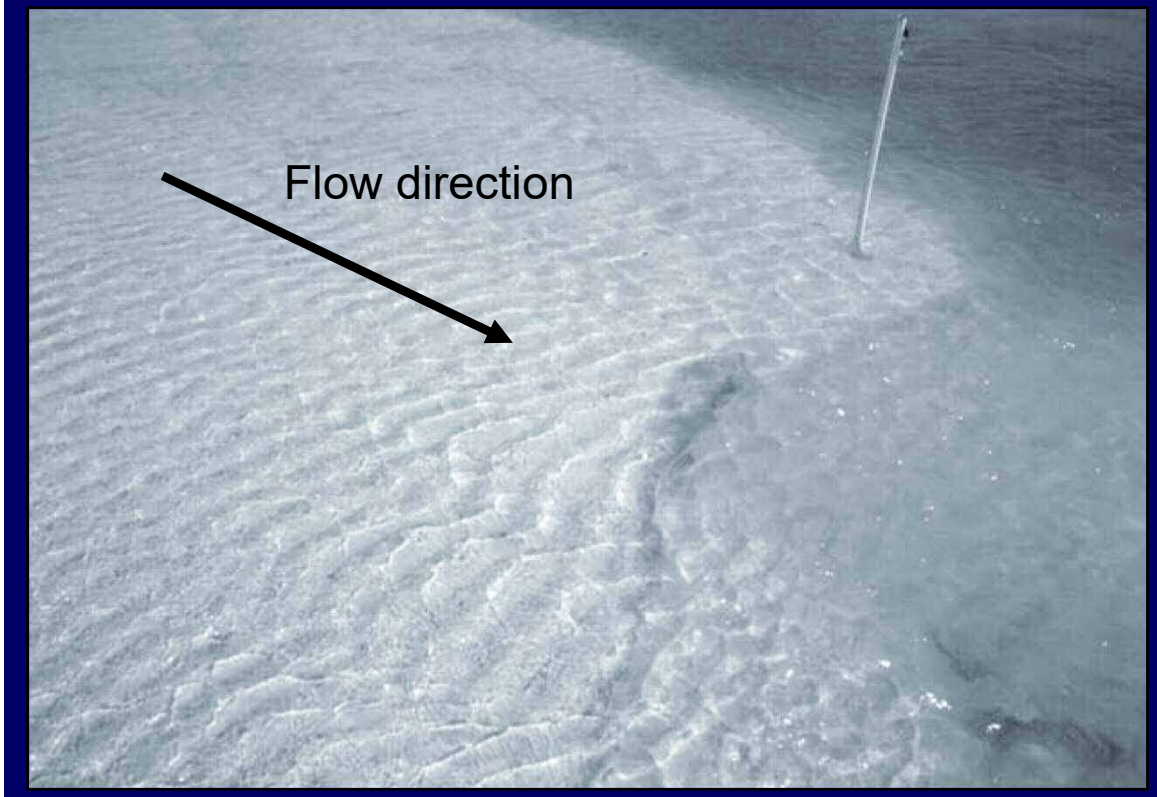


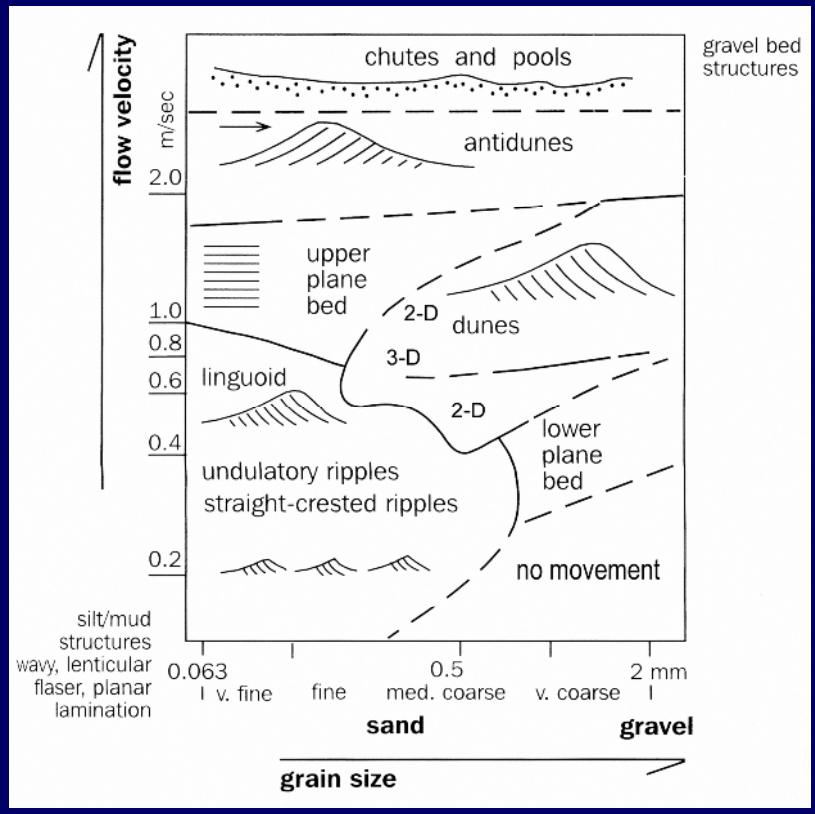
Bedform stability diagram for unidirectional flows.
Mean flow depth 0.25-0.4m

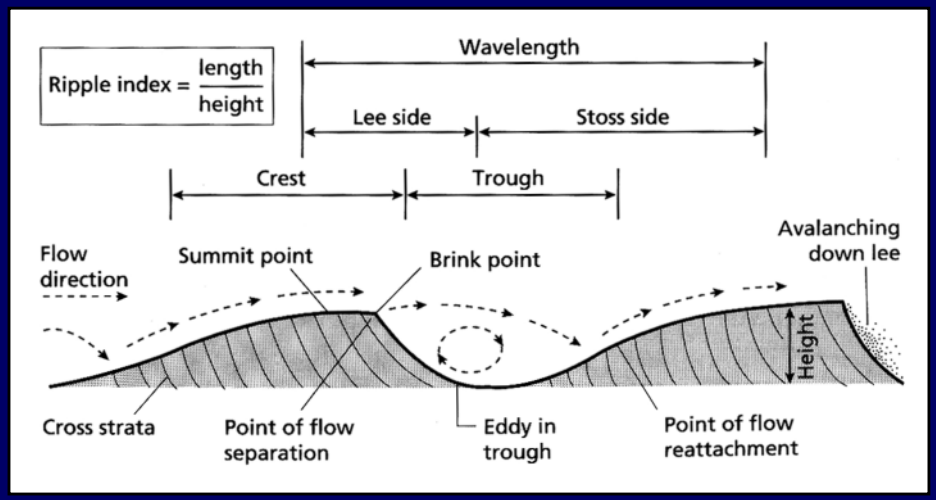
(ii) Oscillatory flow cross stratification

- Small and large scale (formed by small 2-D and large 3-D ripples (no dune stability field)

Small Scale Cross Stratification: Current ripples





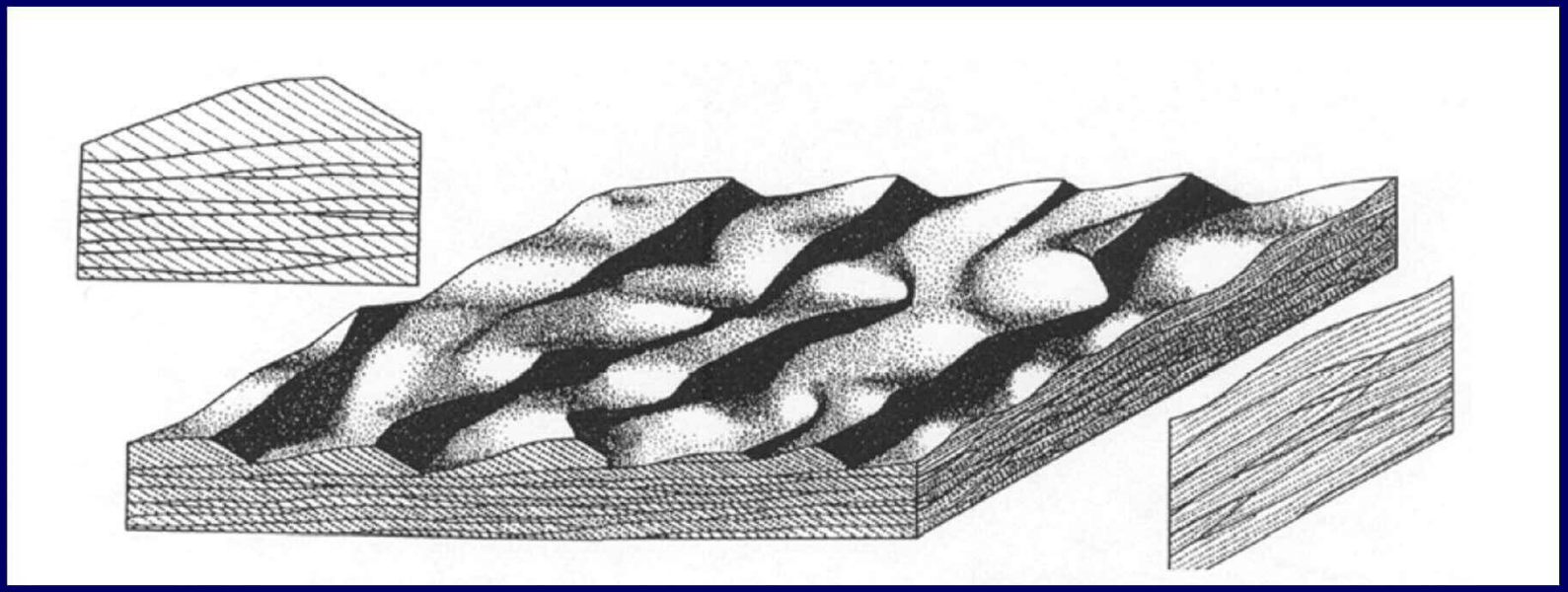


Ripple terminology and flow patterns

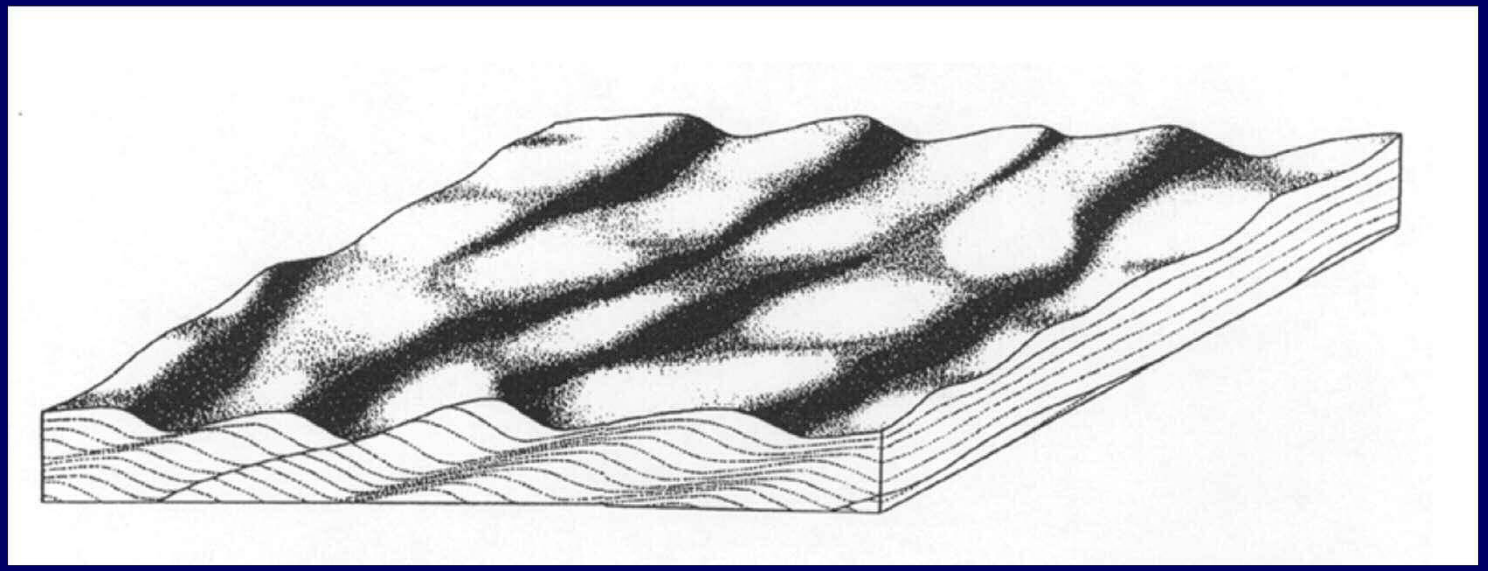
The preserved height of the sets is seldom greater than 2-3 cm (because it's always some fraction of the ripple height)

Ripple height typically 2-3 cm, max height 5-6 cm

Small Scale Cross Stratification: Current ripples



Small angles of climb



Large angles of climb

Block diagrams showing geometry of climbing-ripple cross-lamination produced at different angles of climb

Large Scale Cross Stratification in Unidirectional Flows

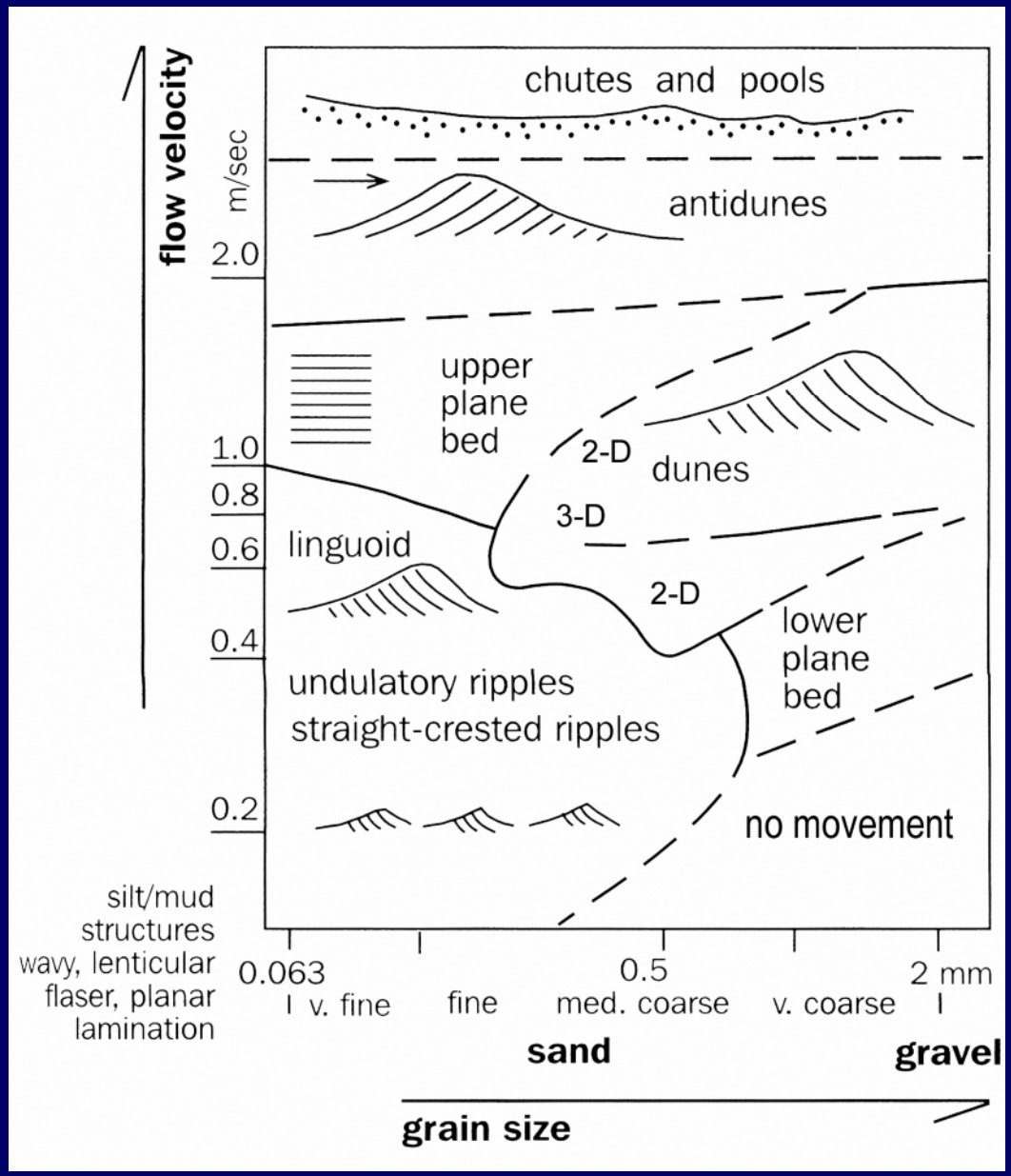
Formed by downstream movement of dunes

Set thicknesses range from <10 cm (arbitrary ~6cm min. cut off) to as much as a few meters

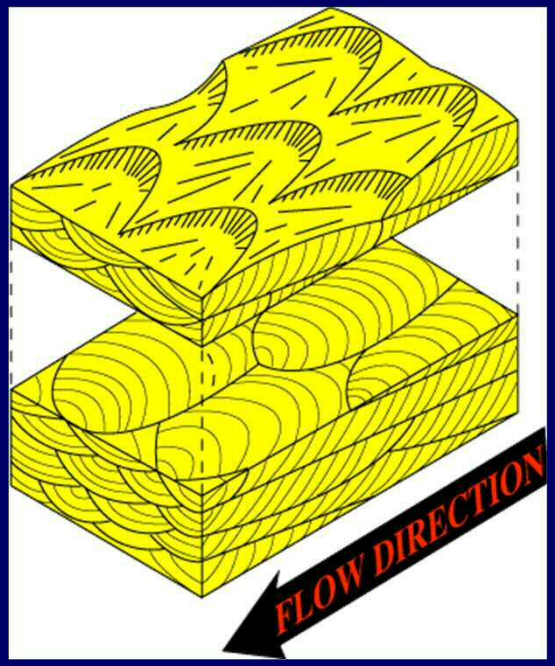
Remember that dunes:

- at relatively low flow velocities have a tendency to be two-dimensional
- at relatively high velocities are moderately to strongly three-dimensional
- at highest velocities wash out to 2-D

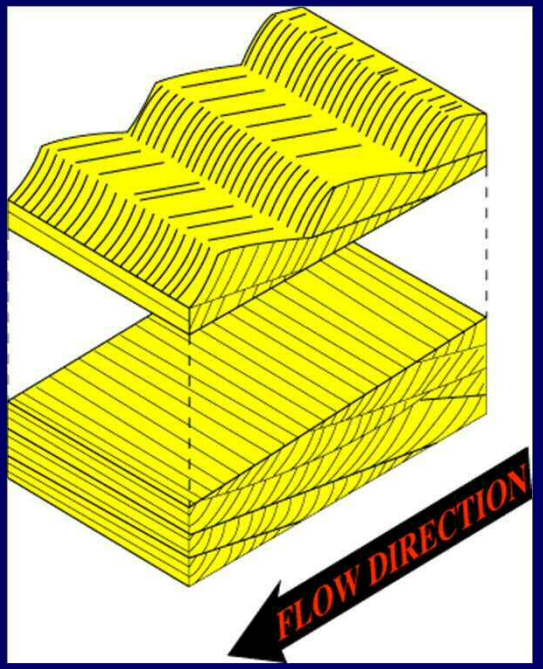
Geometry of cross-bedding varies greatly and depends on whether dunes are 2-D or 3-D



Influence of bedform geometry on geometry of sedimentary structure



Trough cross stratification formed by 3D ripples or dunes



Planar, tabular cross stratification formed by 2D ripples or dunes

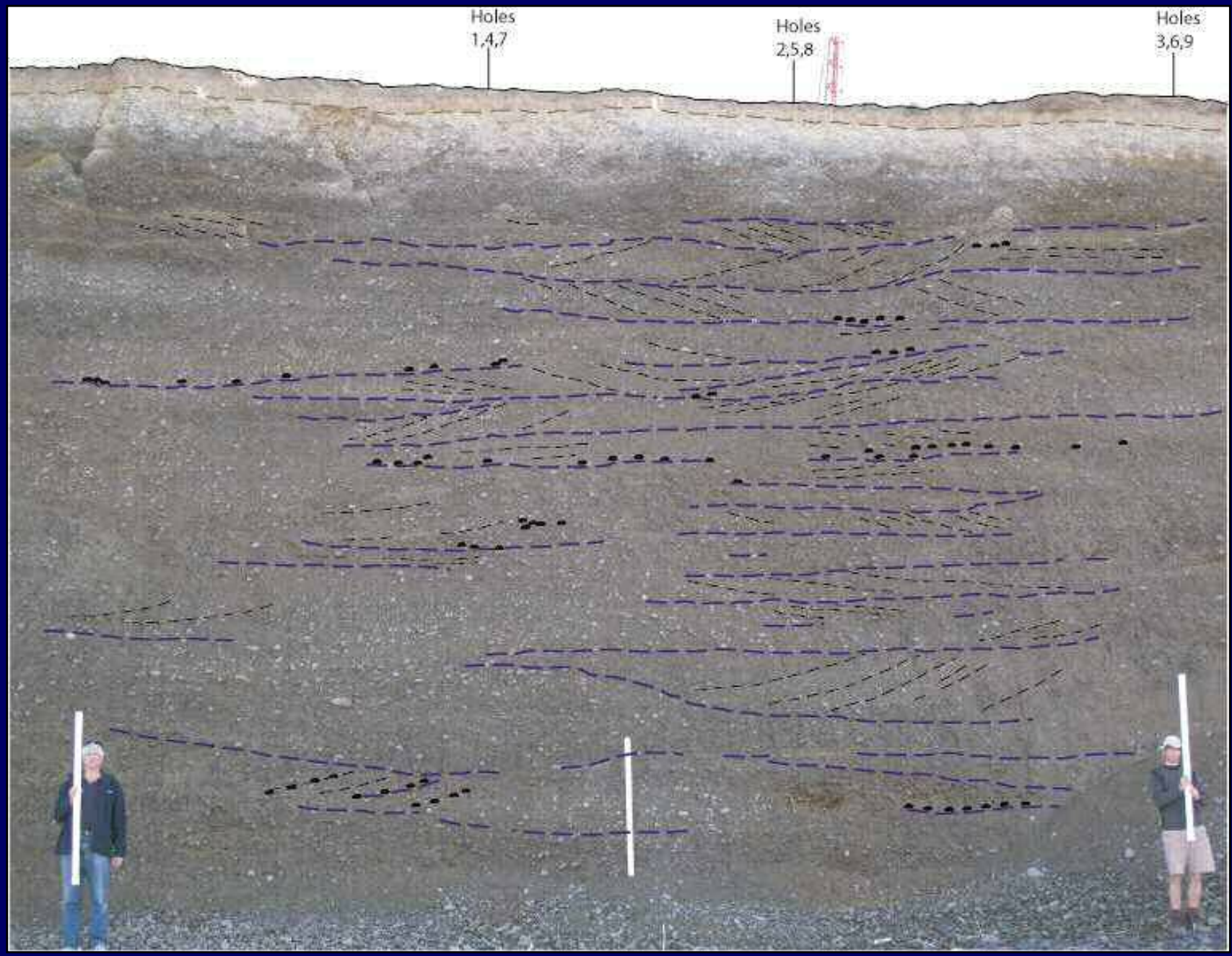
Migrating straight-crested ripples less common form for ripples as mature ripples are 3-D- revisit Class 3 for interpretation

Dunes sometimes climb at higher angles, but less commonly that ripples (because it's uncommon for fairly coarse sediment to be settling abundantly out of suspension over large areas to build up the bed rapidly)



Cross-bedding in coarse-grained sandstones- 2D or 3-D Dune?

- formed by migration of a dune down current
- height used to determine flow depth- what do you estimate?
- cross-bedding dips to left shows palaeotransport direction and way-up of beds

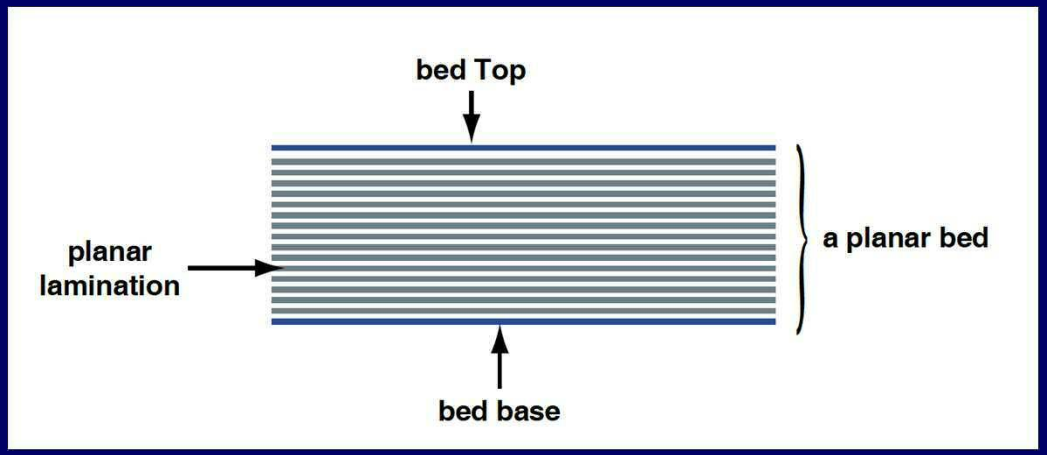


Cross-bedding also occurs in conglomerates

Planar Stratification

Geometry inherently simpler than cross stratification
Occurs in conglomerates, sandstones, mudstones and limestones

Planar stratification: distinction between planar bedding and planar lamination



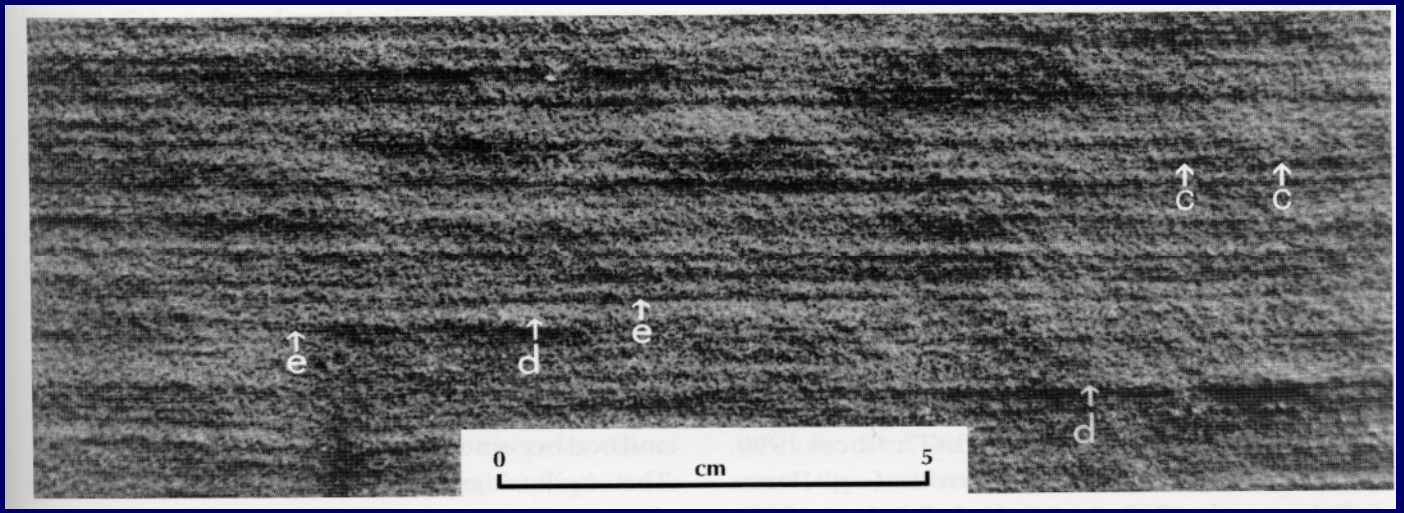


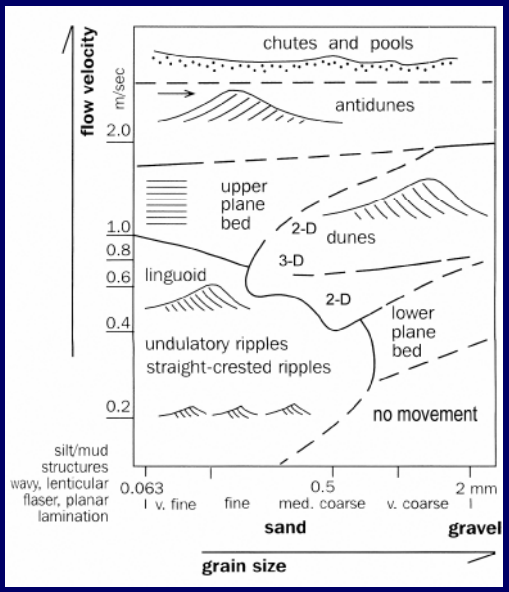
Origin of planar lamination

- Settling from flow
- Downstream movement of very low-amplitude bedforms (sandwaves) on an almost planar surface (e.g., upper plane bed)



Upper Plane Beds



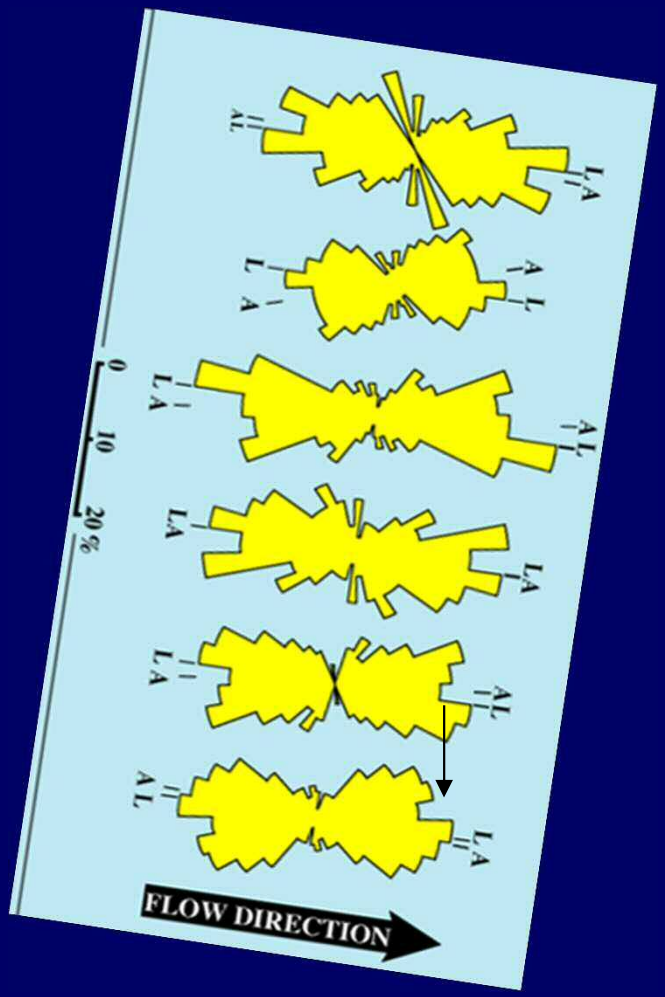


Upper flow regime planar lamina not really planar-result of fast-moving, low-relief (few mm), long wavelength, lee-faceless sand waves

- any grain size at very high shear stresses (upper regime)
- fundamental condition for producing upper plane bed is dominance of transport in suspension
- very high sediment concentrations (high fallout rates)
- sediment transport as a sheet of moving sediment 'traction carpet'
- laminations defined by subtle differences in grain size, particle shape,orientation or composition

Theoretically forms in any grain size but required transport mechanism restricts grain size

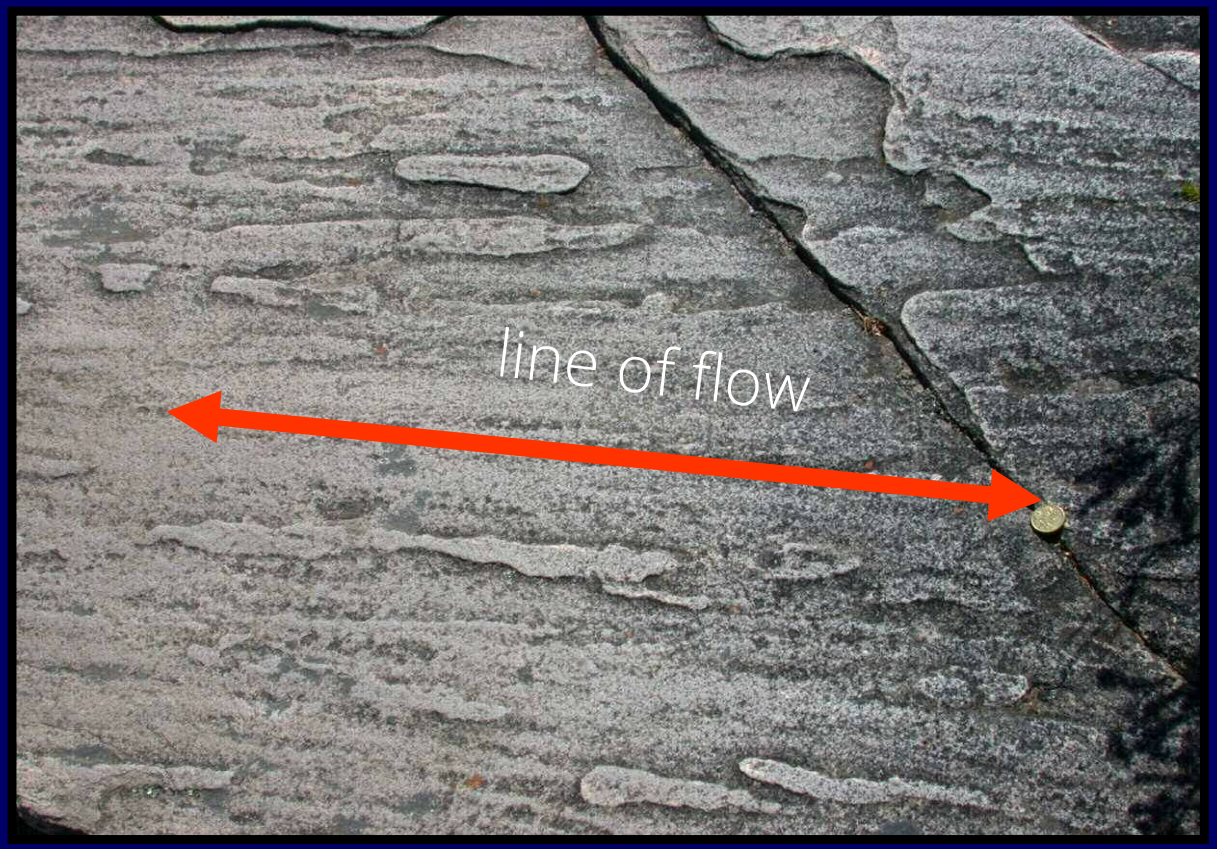
Development of parting lineation on bedding surface of upper plane bed



Vector mean a-axis orientation of sand grains (Cheel, 2006)

Grain long axes are flow parallel

Causes sandstone to break preferentially along that direction



Suggested by Weedman and Slingerland (1985) to form due to streaks in the boundary

Small scale structures of the viscous sublayer: streaks

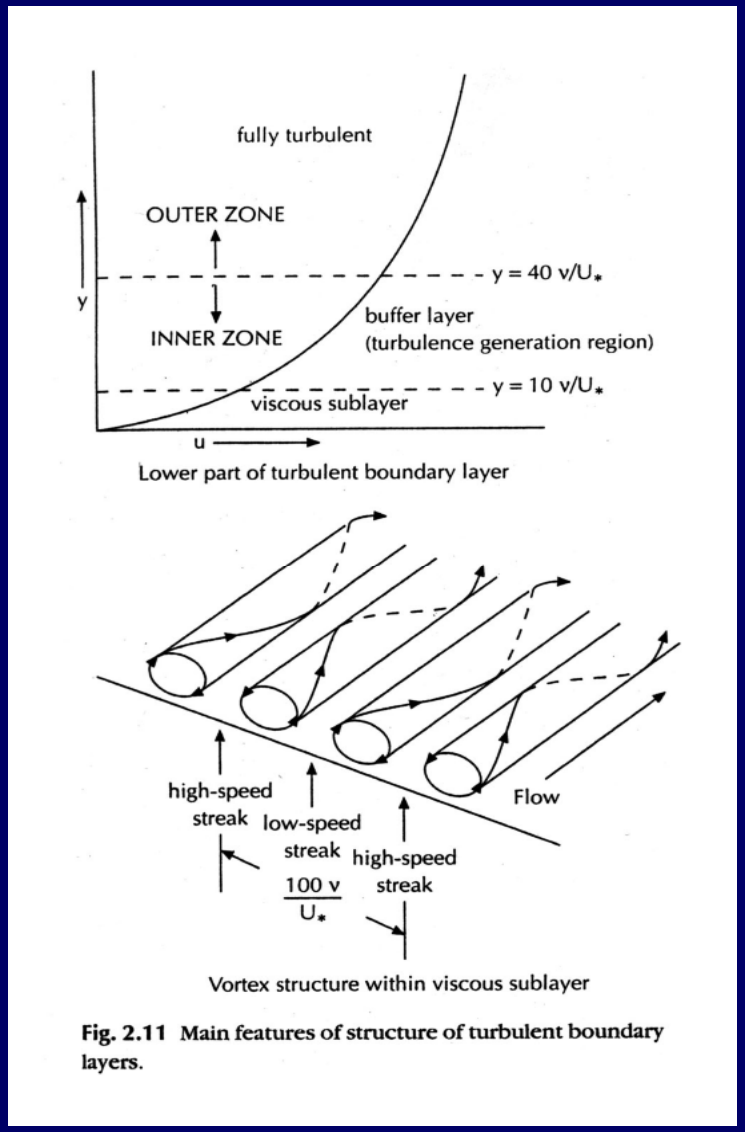
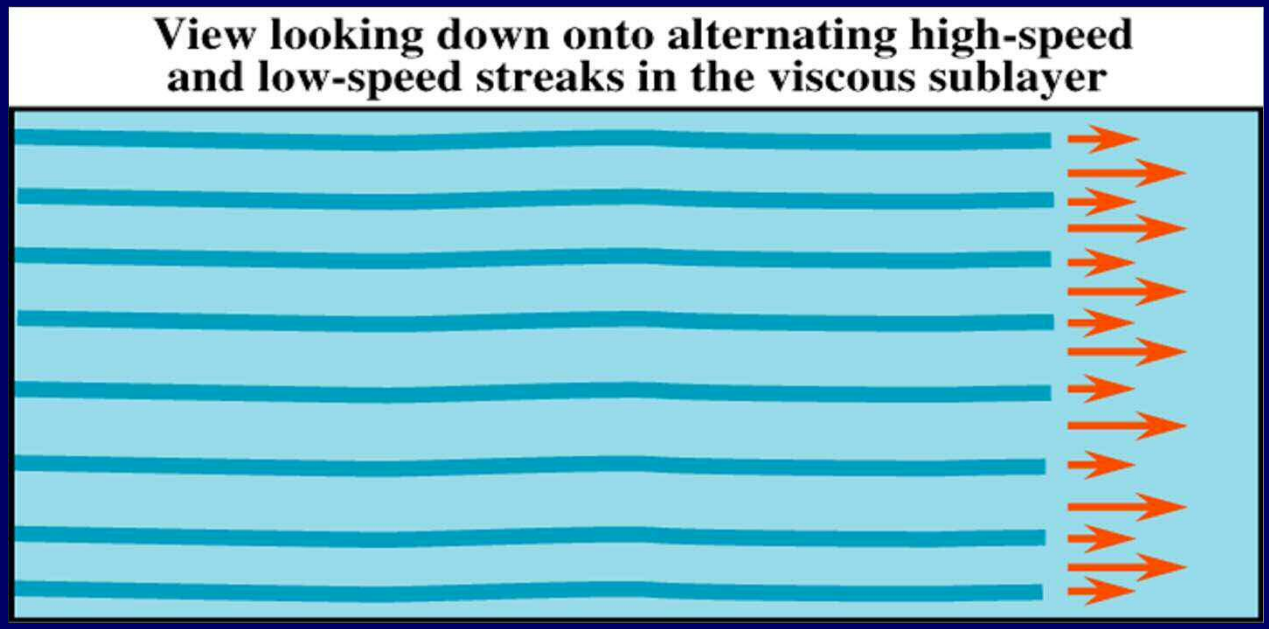


Fig. 2.11 Main features of structure of turbulent boundary layers.



(Cheel, 2006)

Streaks:

- alternating lanes of high and low speed fluid within the VSL.
- associated with counter-rotating, flow parallel vortices within the VSL.
- push grains aside to form tiny ridges parallel to flow ('parting lineation')
- parting lineation valuable for determining palaeoflow direction

Small Scale Cross Stratification: Wave ripples



- symmetrical in cross-section, continuous (bifurcating) crest lines, broad troughs and peaked crests
- ripple index 6-10

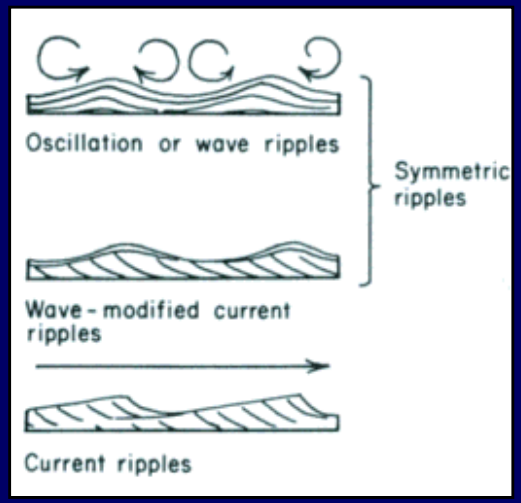


Wave ripples can form in much coarser sands than current ripples, and can get much larger (wave ripples in Wales)





Wave formed ripples on tidal flats may be asymmetric because the two flow directions are not of equal strength/duration





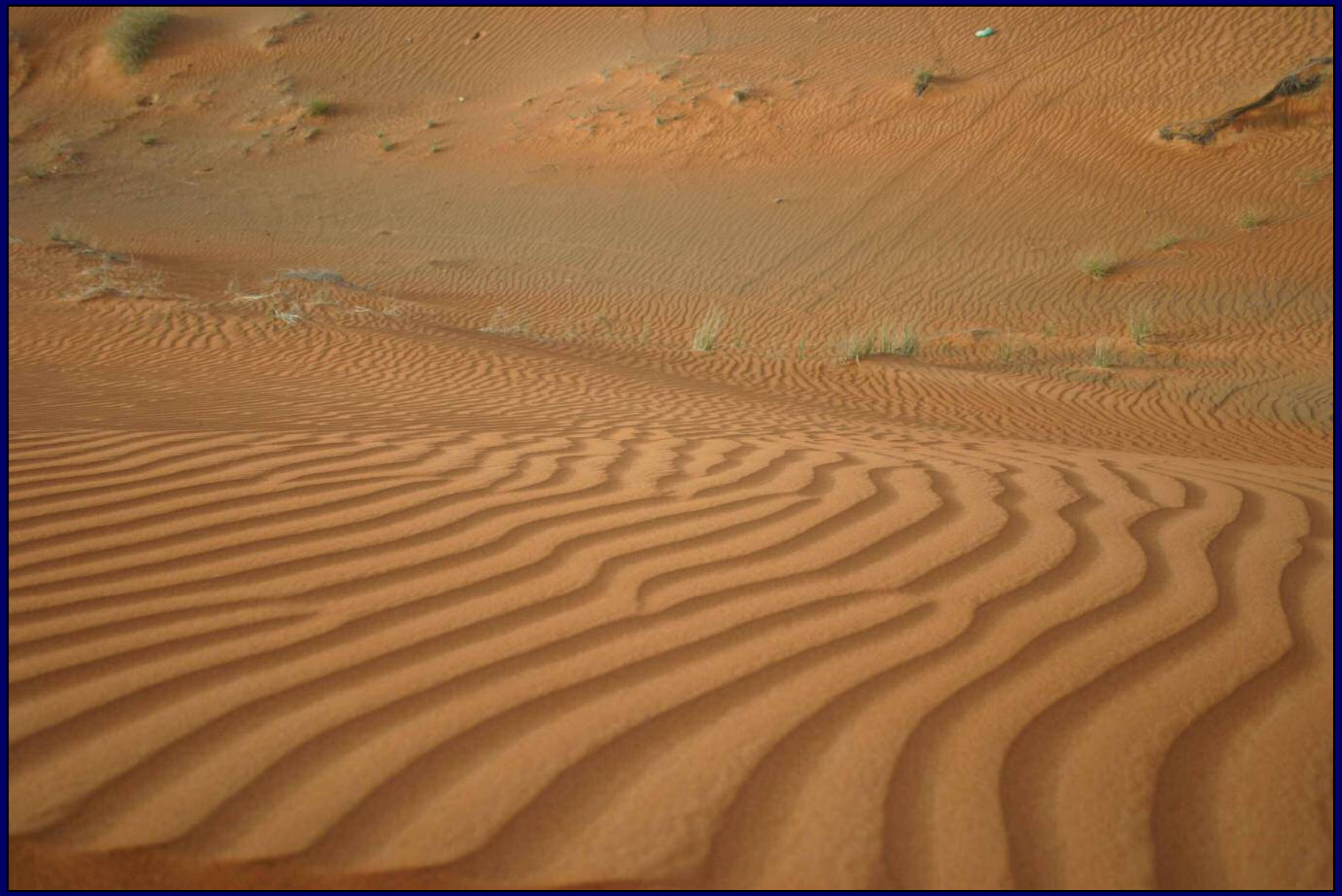
Modern beach, Phillip Island, Victoria
Ebb and flood tide ripples superimposed



Mezo-Proterozoic, Western Australia
Small ripples formed in troughs of larger ones at low water

Complex interference patterns can form in areas subject to tides

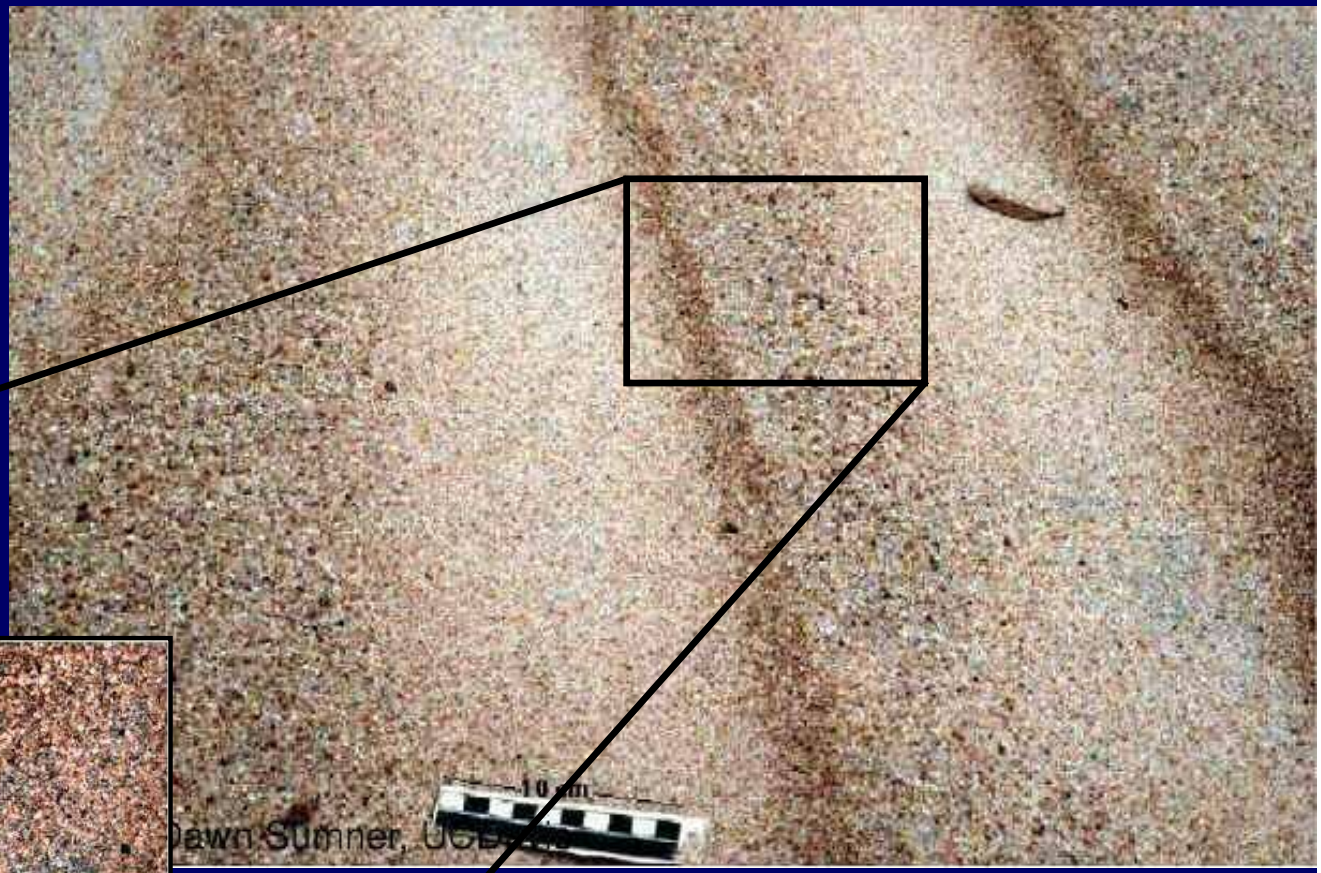
Small Scale Cross Stratification: Wind ripples



Ripples crest straight to slightly sinuous, very persistent, may bifurcate

Grain size segregation associated with wind ripples

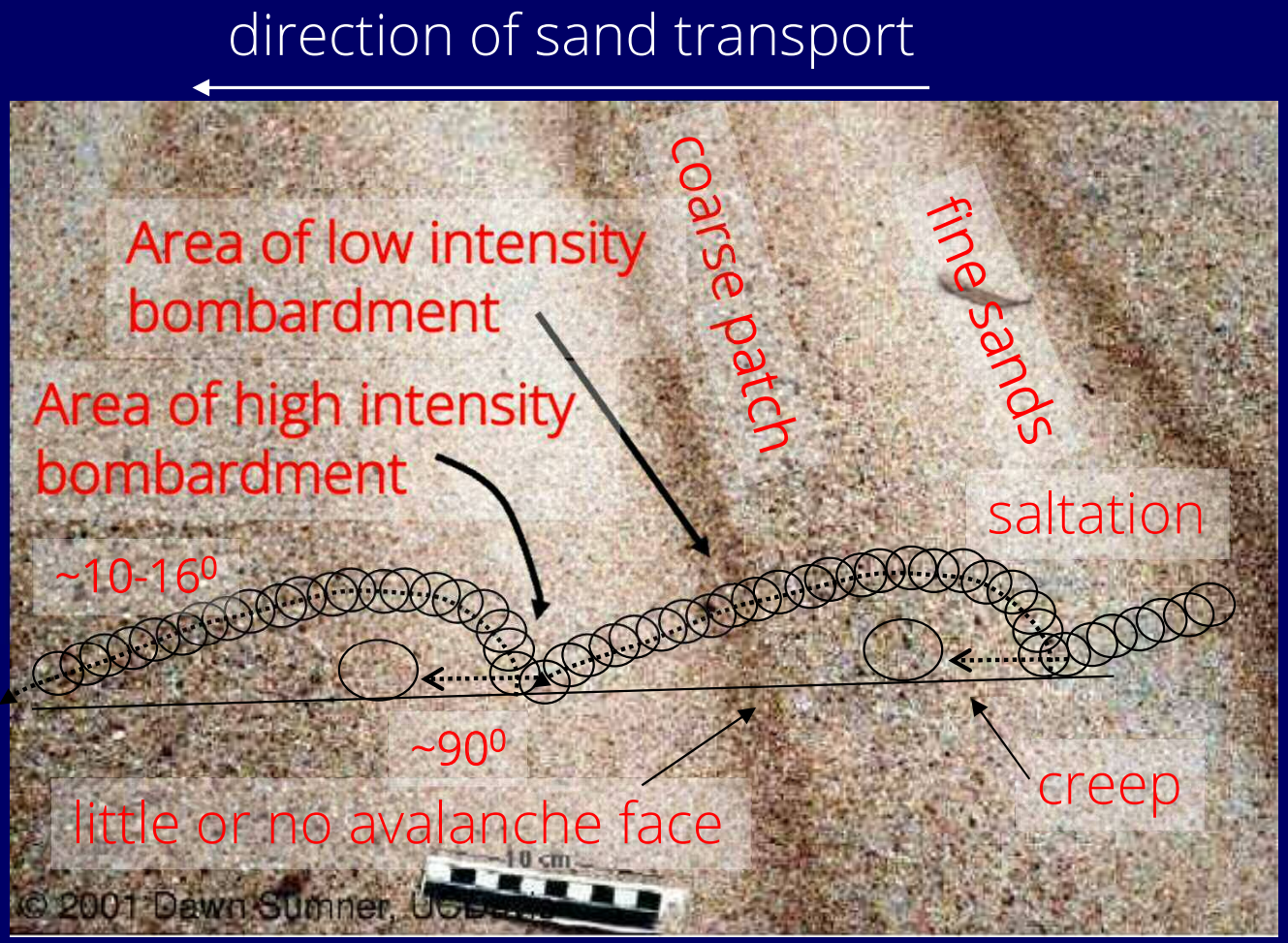
Modern aeolian ripples
Brazil





- asymmetrical in cross-section
- coarser grains on crest, finer grains in troughs

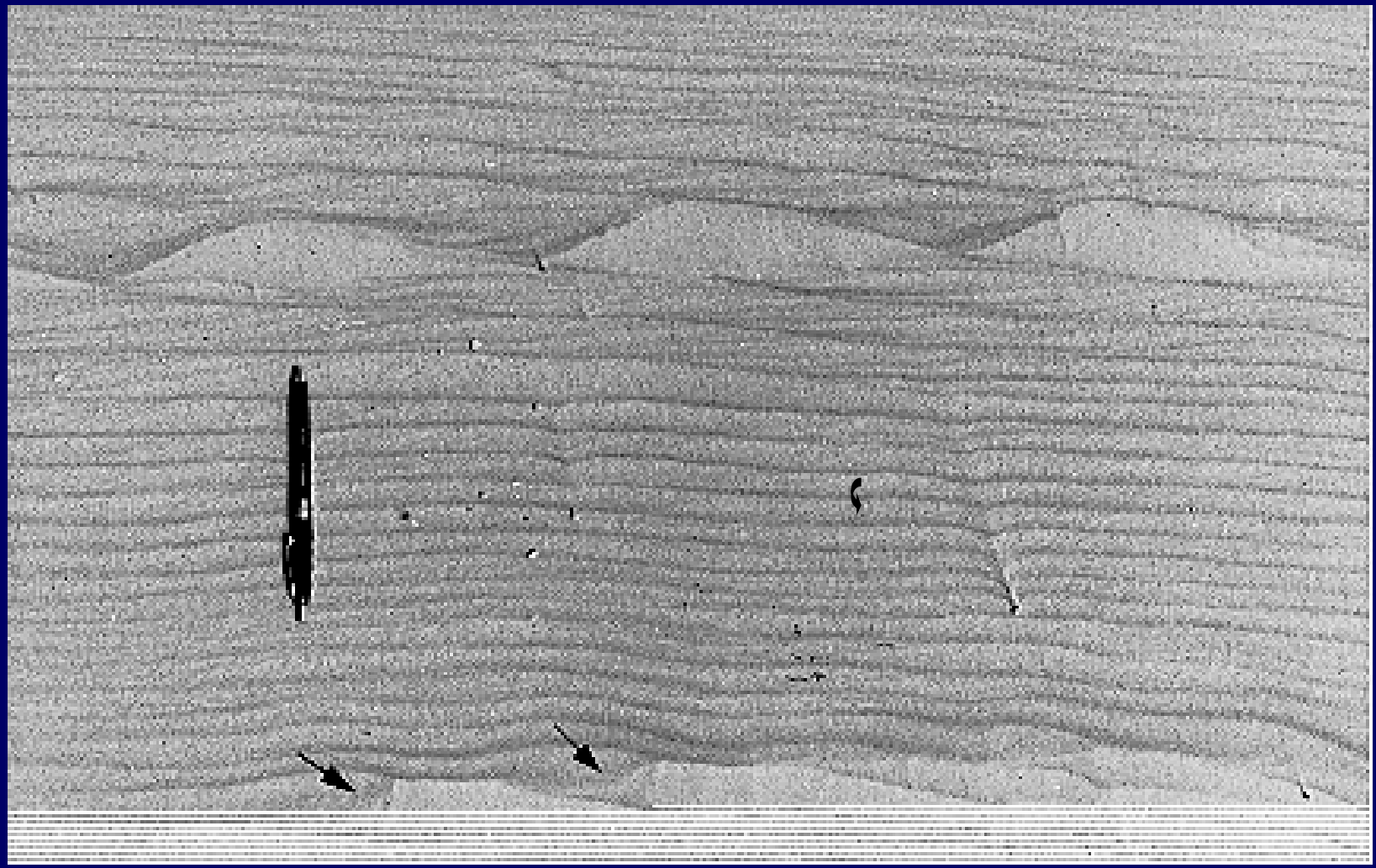
Wind ripple migration processes



- kinetic energy of the impacting grain > energy necessary to raise a similar grain out of its pocket in the bed, against the force of gravity
- ripple migration dominated by grain saltation not repeated lee-side avalanches

Internal structure of ripples

wind direction



- cross-laminae/foresets frequently only weakly developed
- strong inverse grading or bimodality of grain size due to concentration of coarser creepload along the ripple crests



Large bedforms – wind-formed dunes

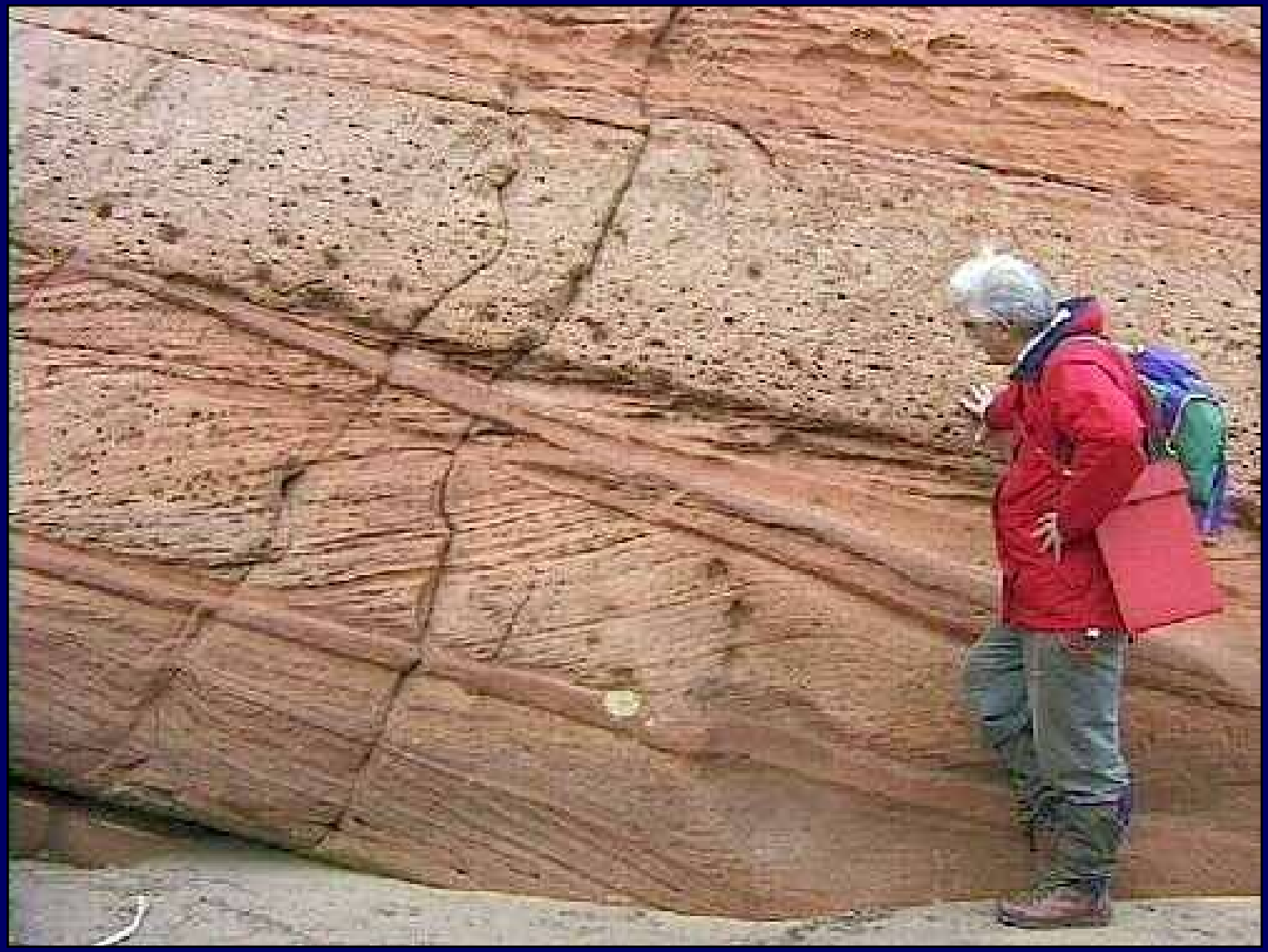
Note smaller bedforms are present on the up current faces of the dunes



Trough cross bedding formed by migration of wind blown (aeolian) dunes

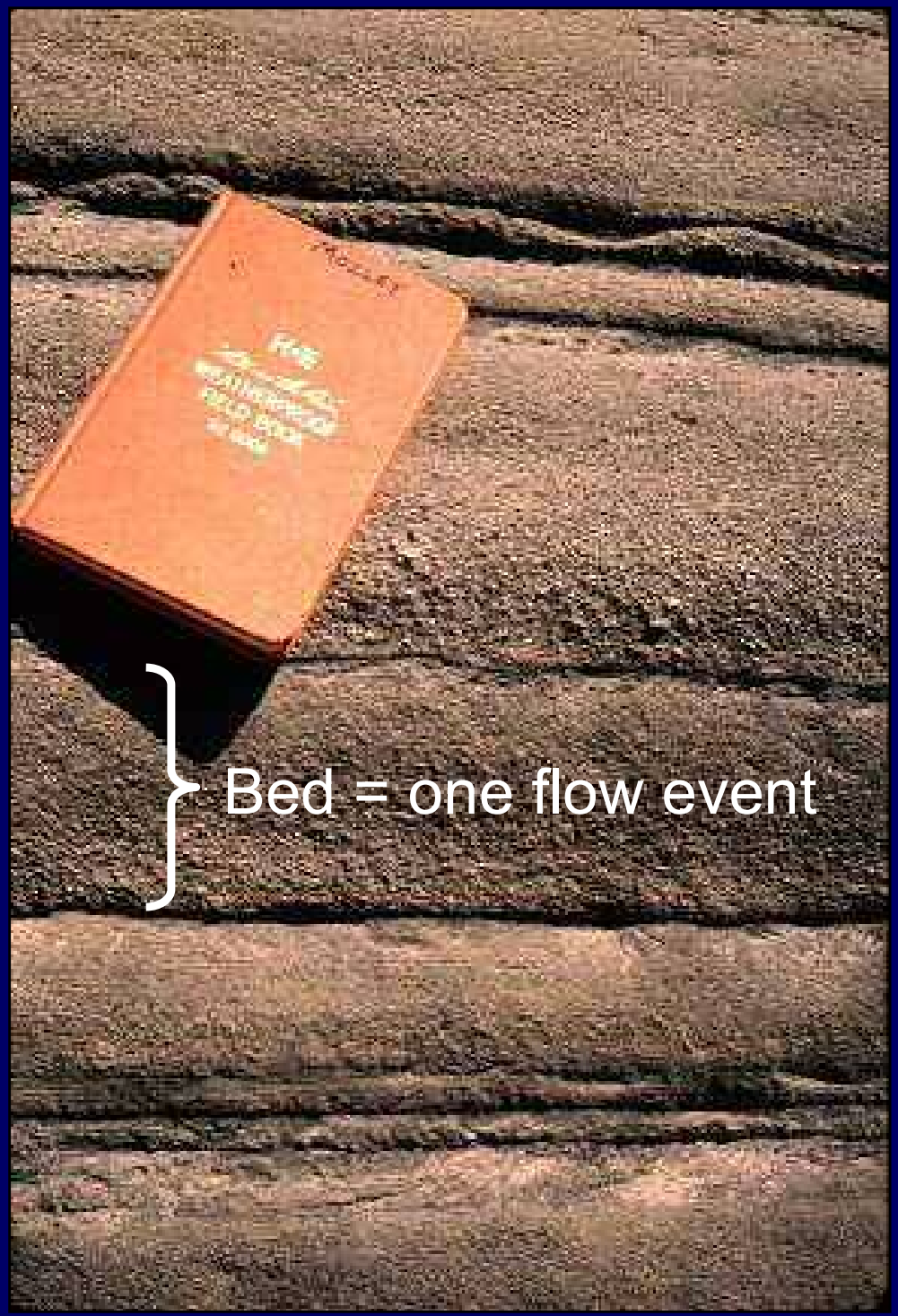
Cross-bedding caused by changes in grain size

Way up determination from cross stratification



Cross-bedding in red sandstones

- tectonic dip is to the right



Graded bedding

Grading: vertical change in grain size (or composition) through the bed
- 'normal' vs 'reverse' grading

Origin of normal grading
When a flow slows down rapidly:
- all the grains settle to the bed
- big grains reach bed before little grains
- produces 'normal grading'
- way-up indicator

Graded bedding in ~10cm-thick beds
Grading is from coarse sand at base to fine sand at top



Graded bedding in finer grained sands and muds is often evident from colour changes

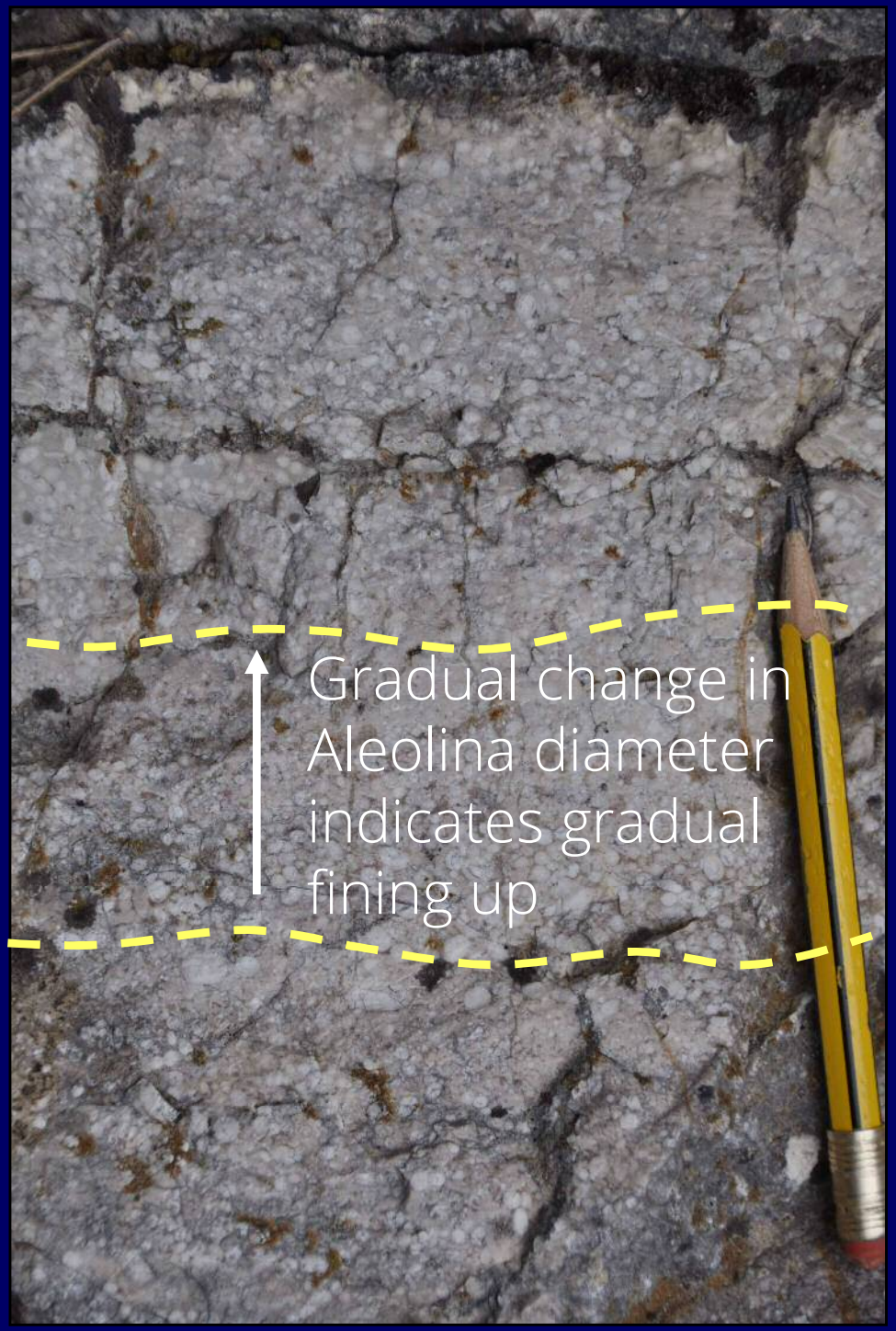
Gradual colour change from pale to dark indicates gradual fining up

The sharp change from dark to light indicates sudden increase in current strength

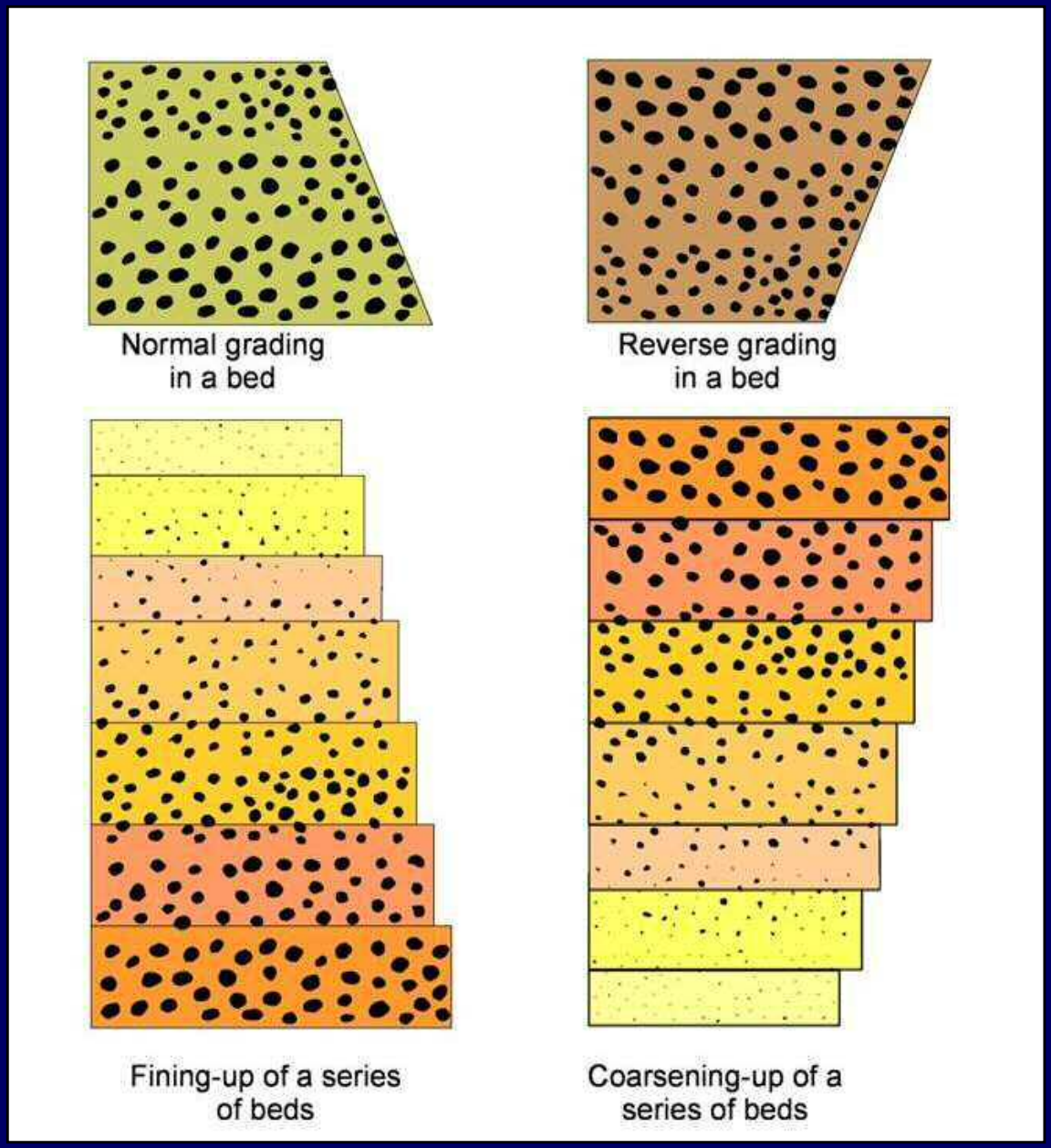
Graded bedding in coarse grained calc-arenites

The grains are composed of single-celled organisms called Alveolina (a foraminifera)





Difference between grading and fining/coarsening



(Nicholls, 2009)

Grading:
Trend in grain size within a single bed

Fining and Coarsening:
Trend in grain size that occurs over a number of beds (i.e. within a unit)

Note:
Individual beds may be normally graded within a coarsening upward succession

Other syn-depositional intrabed structures/textures

- Massive beds
- Imbrication
- Sole marks
  - Scour marks (including flute casts)
  - Tools marks

Massive Beds

Defined by an apparent absence of any form of sedimentary structure

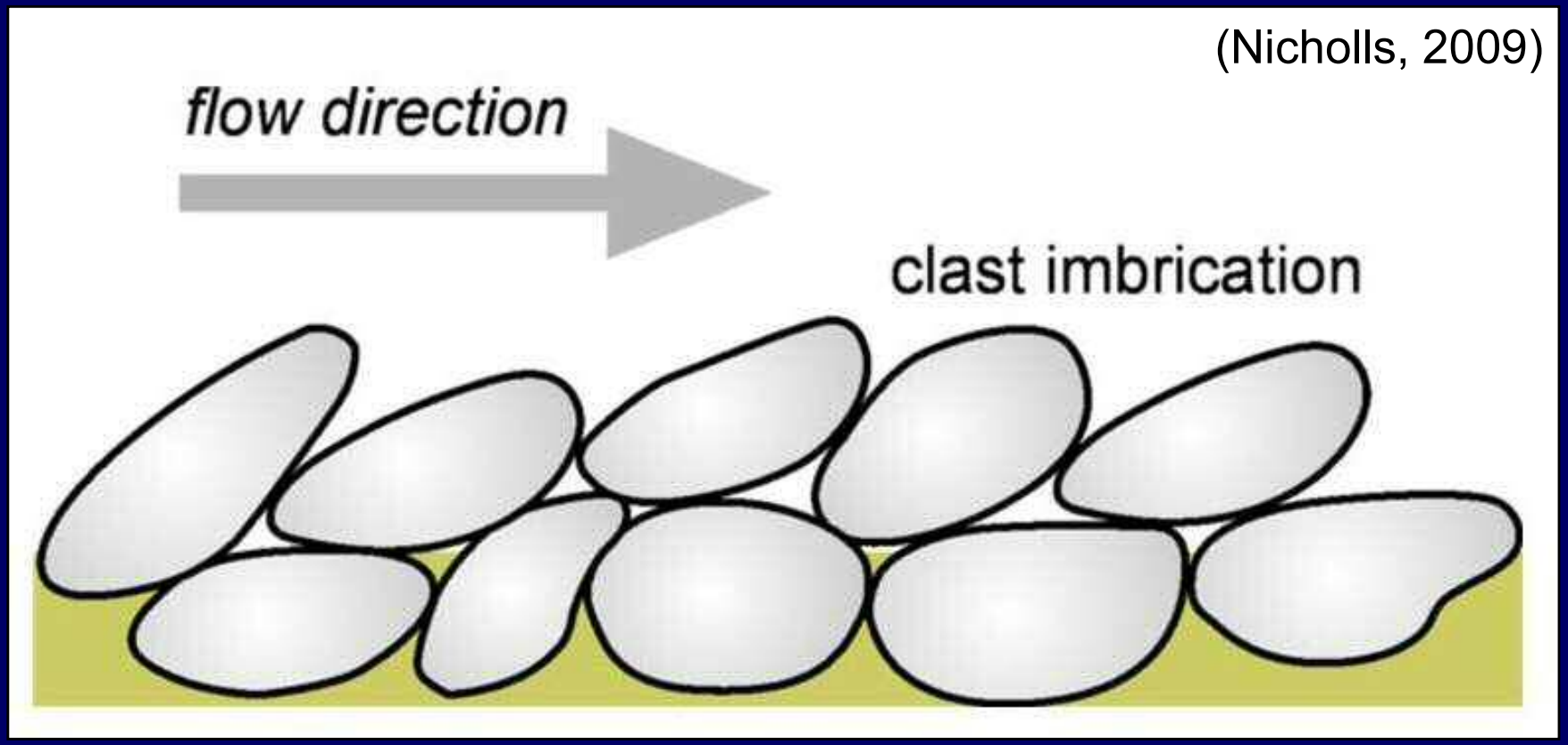
Variety of causes:

- diagenesis (e.g., limestones and dolomites that have been recrystallised)
- intense bioturbation
- very low-energy environments that produce claystones, chalks and marls
- very well sorted sandstones where the structures are not delineated by textural variations apparent to the naked eye. Layering may be evident in thin section



Imbrication of grains

- not strictly a structure – more of a texture but frequently seen in conglomerates
- also occurs in sands but too small to see in hand specimen



- elongate pebbles stacked up, dipping upcurrent
- indicator of current direction



Pebble imbrication in a river gravel
Pebbles dip upstream, so current flow here is from right to left



Conglomerate showing pebble imbrication

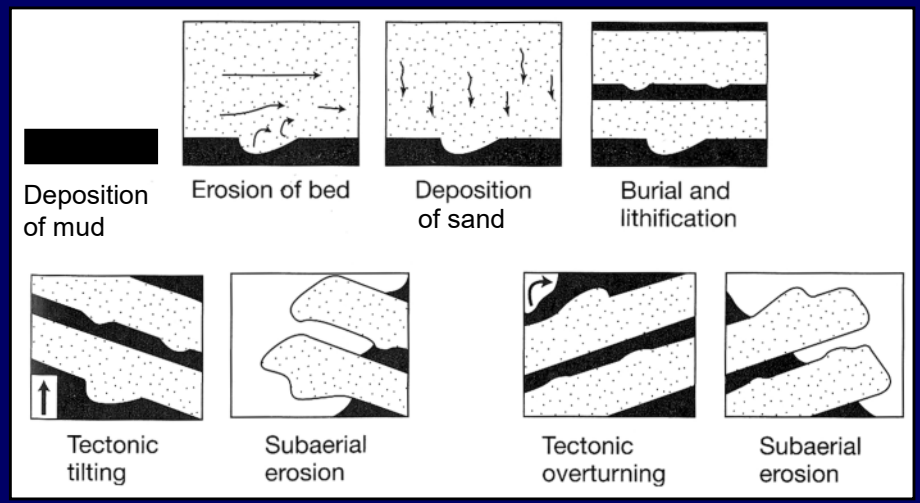


Sole Marks

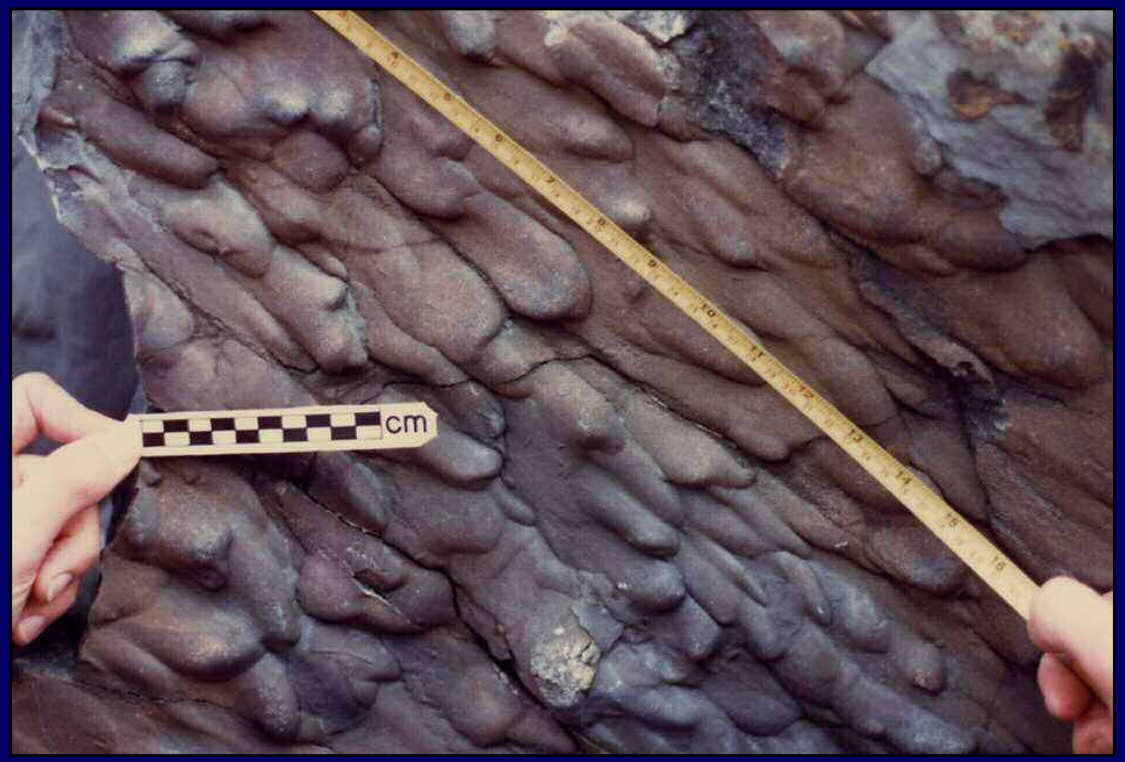
- Sole marks: a group of sedimentary structures normally seen on the underside of beds in positive relief
- A common feature of successions of alternating sandstones and mudstones with the structures seen on the bases of the sandstones
- Result from a short-lived current acting on a cohesive or semi-cohesive bed, usually fine grained sediment (mud)
- Two broad classes:
- Scour marks: caused by erosion (turbulent scour) of the bed by a strong current (include obstacle scours, longitudinal scours and gutters and flute casts (aka flute marks)
- Tool marks: caused by mechanical disruption of the bed by objects carried by a strong current

You almost always see sole marks in negative relief

Useful as paleocurrent indicators and way-up



Flute marks

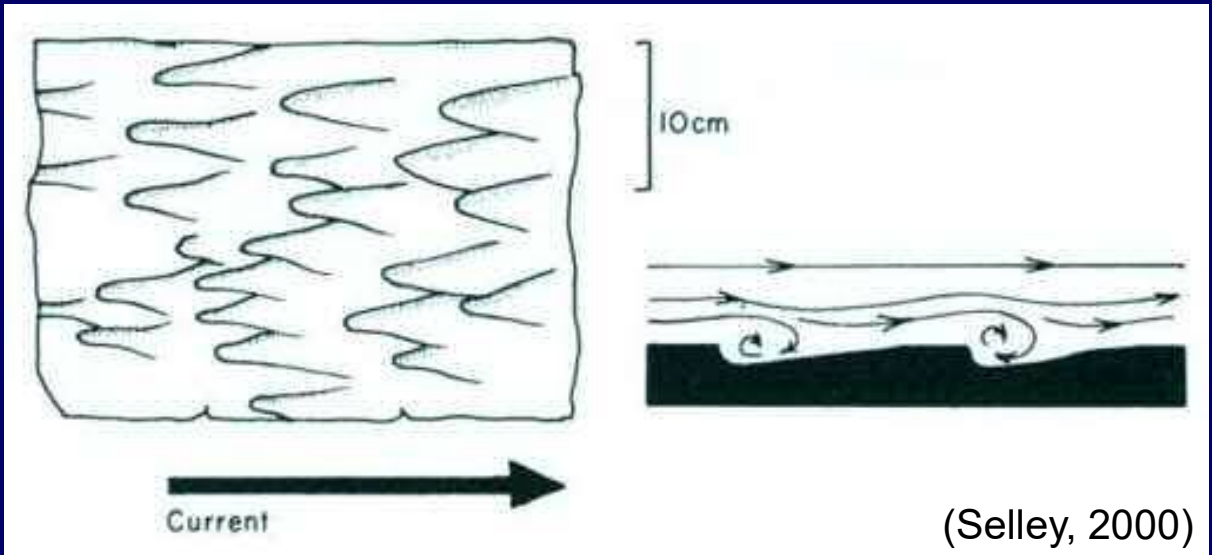


Flute marks viewed on underside of bed

- heel-shaped hollow, scoured into mud bottoms
- infilled by sand, contiguous with overlying bed
- 1-5 cm wide
- 5-20 cm long but can get giants a few meters across!
- typically gregarious

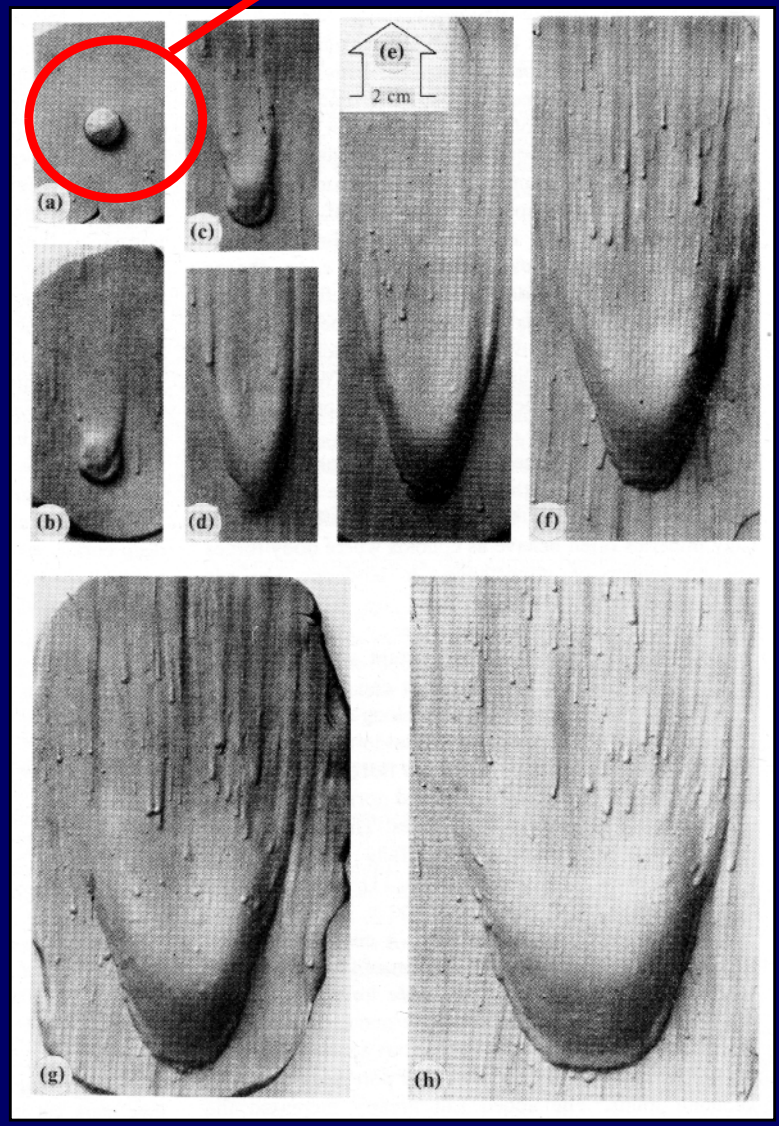


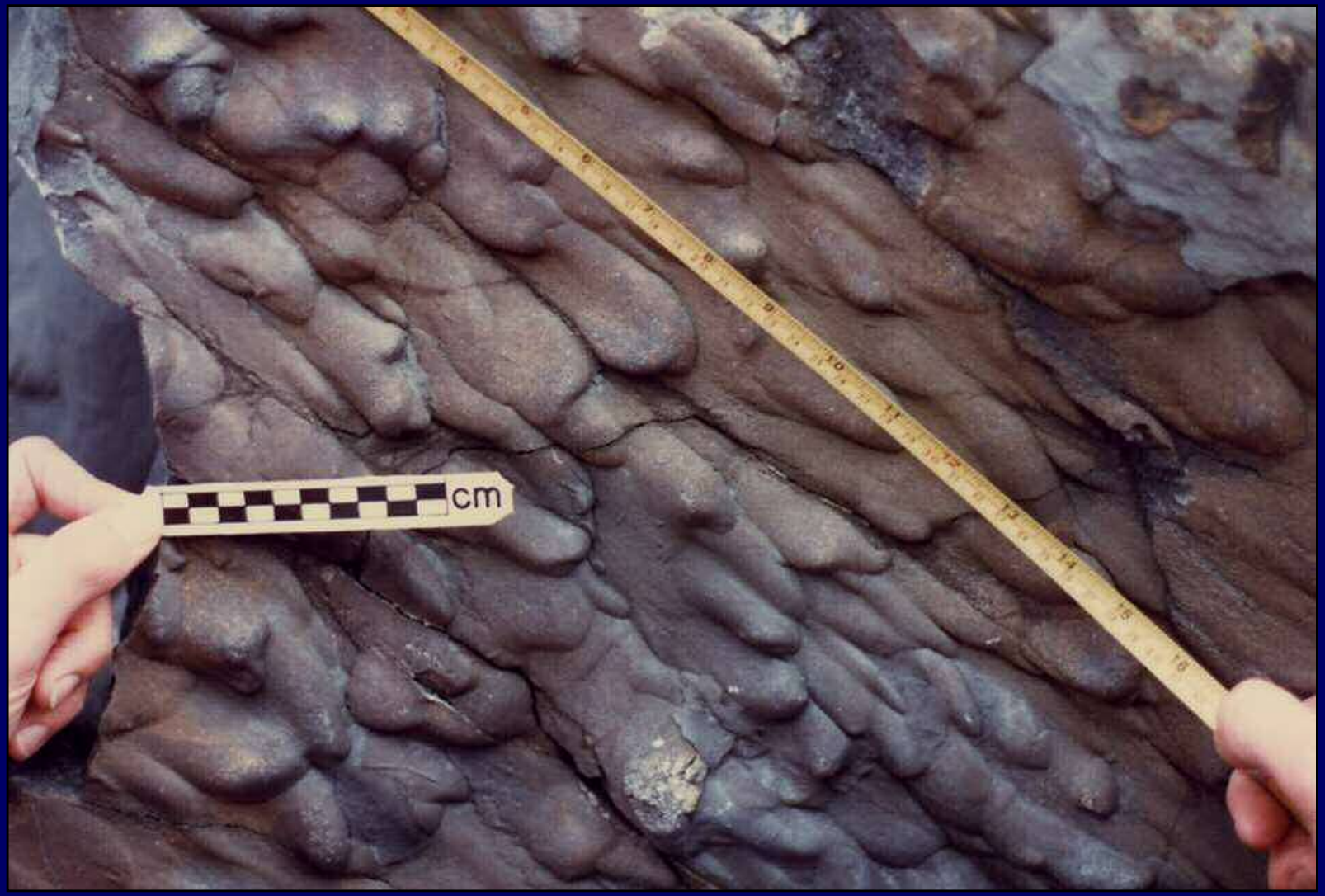
Formation of flute marks



Sketch of flute marks and cross-section showing scouring of fluted hollow in soft mud by current vortices

Allen described experiments that used plaster of Paris (gypsum) in a flume to show the development of a flute from an initial bed defect (e.g., Allen, 1971, 1982)





Flow direction is from bottom right to top left

- rounded part of flute is upcurrent end
- flared end points downcurrent

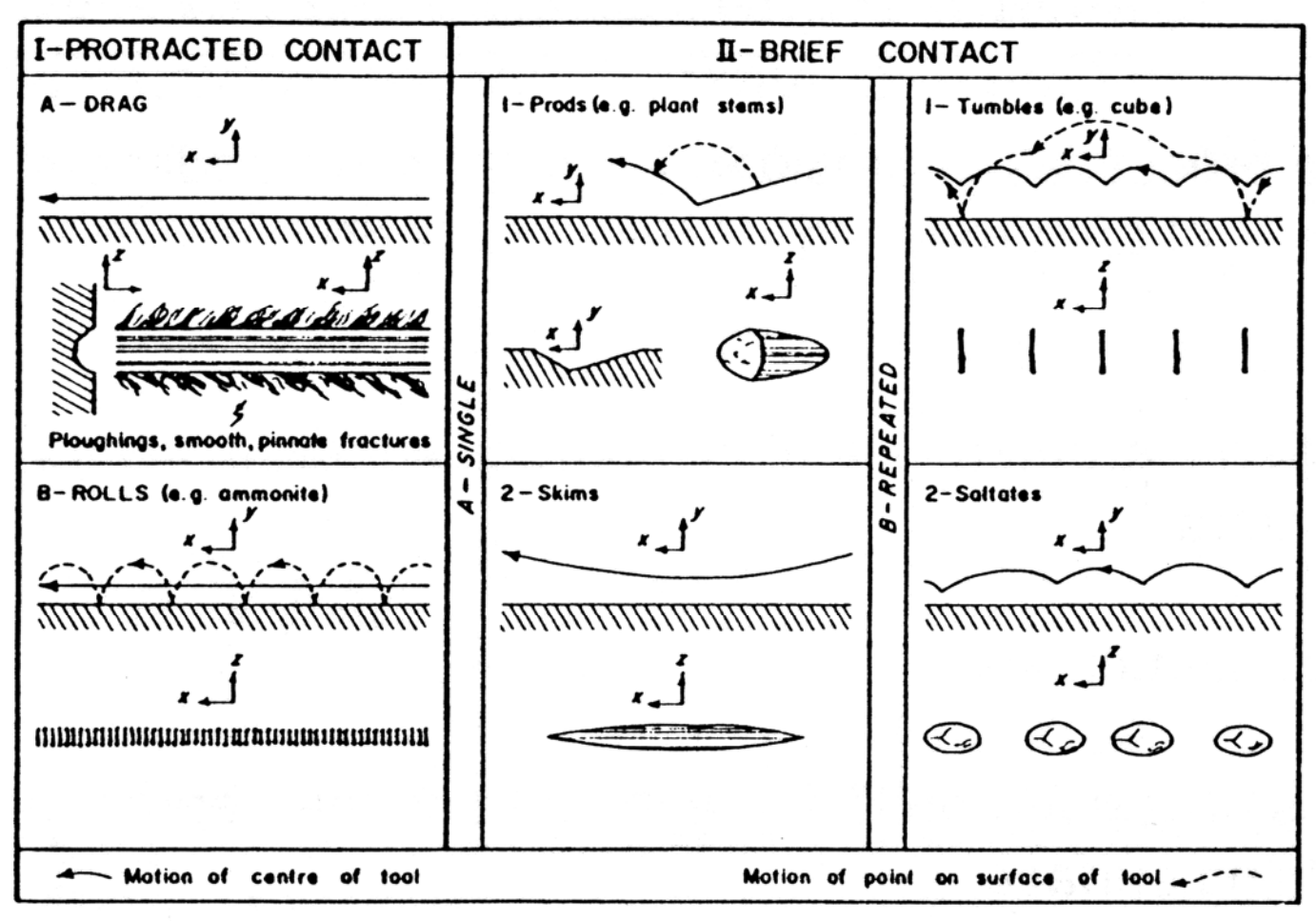
Tool marks

Erosional bottom structures caused by moving objects in the flow

- Irregular in shape
- Roughly oriented parallel with the flow

Types of tools:

- Pebbles (incl. mud clasts eroded from the bottom)
- Wood and plant fragments
- Shells
- Fish vertebrae



Secondary (Post-depositional) structures

Post-depositional structures formed while sediment is still 'soft', long before lithification

- on scales of centimeters to a few meters

Timing:

- only shortly after burial (while sediment is noncohesive and buried < a few meters
- during, not after, deposition

Examples:

Slumps and slides

Dewatering structures due to liquefaction and fluidization

- convolute bedding, overturned cross-bedding,
- sandstone dykes, mud volcanoes

Load structures

- load casts, ball-and-pillow structures

Dessication

- mudcracks

Biogenic structures

- tracks, trails and burrows

Slumps and slides



Involve penecontemporaneous plastic deformation of sand and mud
Shows evidence of lateral movement
Requires rapid deposition of muddy sediment on unstable slope
Lateral movement initiated by earthquakes, storms or spontaneously

Slumps and slides



How can we tell the deformation is penecontemporaneous?

Dewatering structures ('quicksand structures')

Key deformation mechanisms of unconsolidated, water-saturated sands

- Liquefaction
- Fluidization

Liquefaction

- temporary loss of strength caused by breakdown of grain framework and transfer of grain support to pore fluid
- associated with increase in pore fluid pressure
- may be result of seismic shaking, impacts, groundwater movement, passage of waves
- no net inflow of pore fluid



Overturned cross-bedding - produced by current shear across the top of a liquified cross-bed set.

Dewatering structures ('quicksand structures')

Fluidization

- occurs when interstitial fluid flows upwards through a granular medium (e.g., sand) at a rate where upward-directed component of fluid drag balances or exceeds the particle weight
- requires external supply of fluid



Convolute bedding: caused by the expulsion of pore water from loosely packed sand

Dewatering structures: Sand and mud volcanoes



Water escape structures driven by fluidization

Load structures



Commonly seen on base of sandstones overlying mudstones

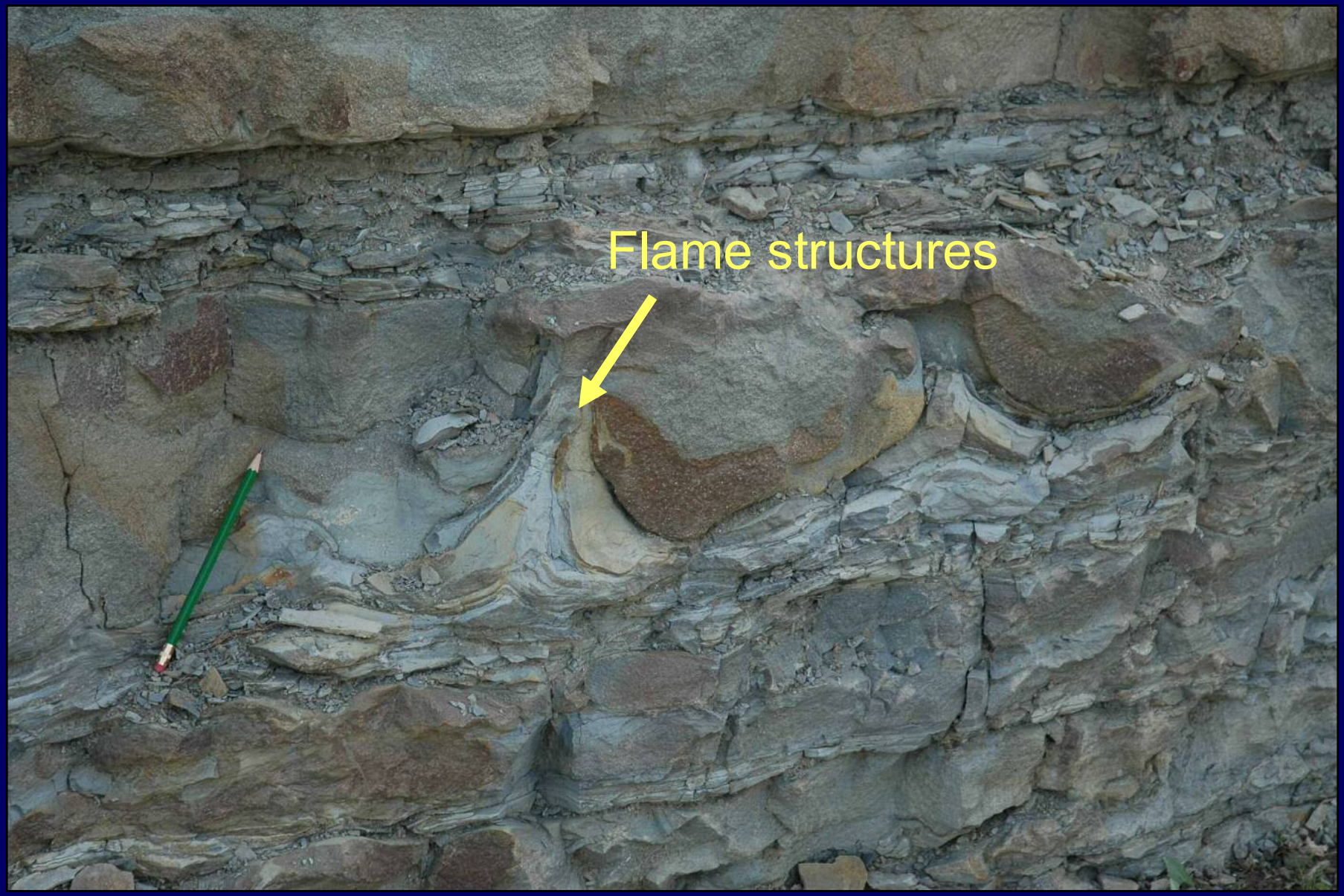
Form due to density contrast between more dense, overlying waterlogged sand and less dense, unconsolidated mud so the sand above has sunk down into the mud beneath



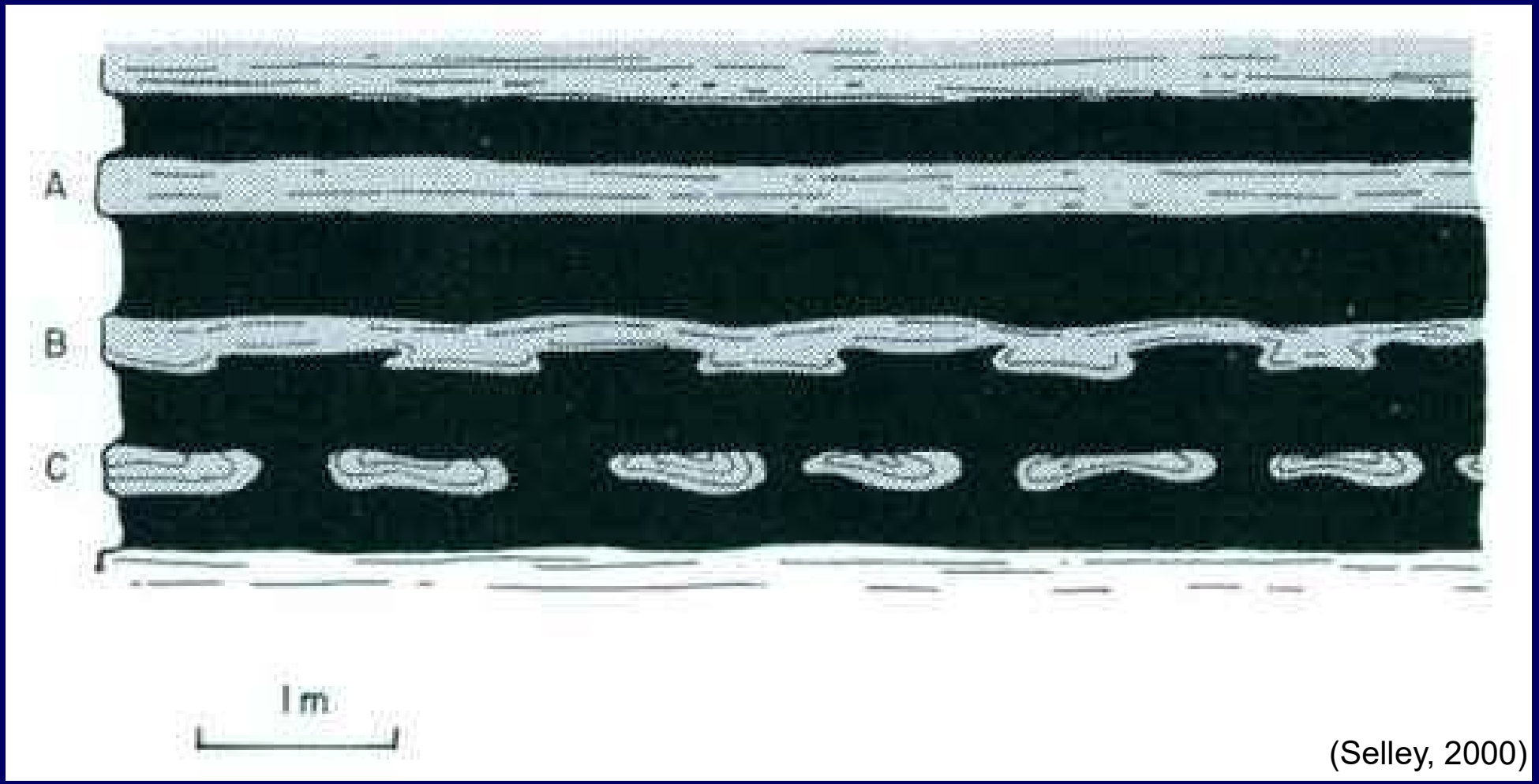
Variation in size and shape: small load casts on the base of a sandstone bed



Variation in size and shape: lobate load casts on the base of a sandstone bed



Dewatering and soft-sediment deformation structures are often associated, because upward flow of water loosens the packing of sediment grains, allowing fluidization to occur

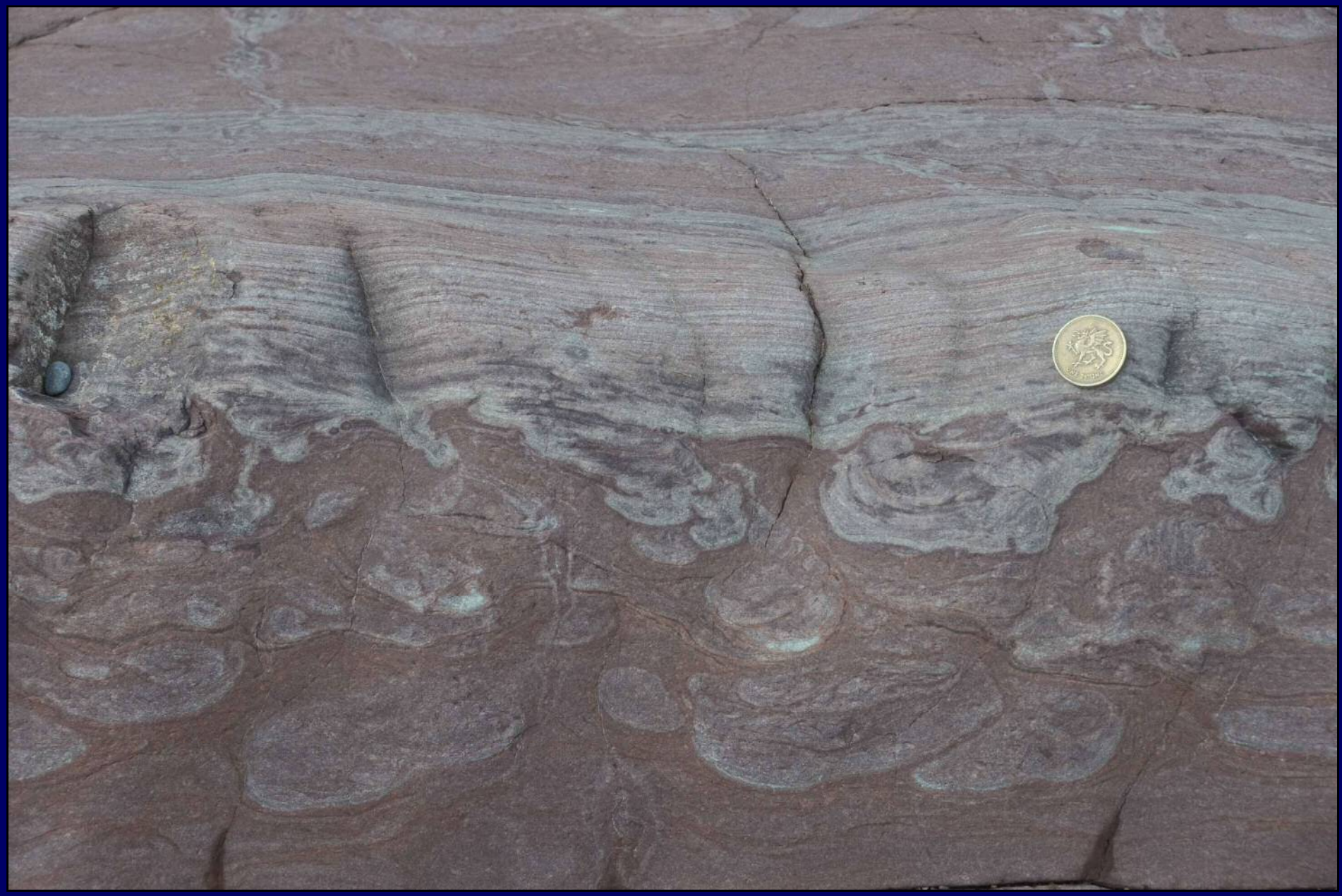


(Selley, 2000)

Bed A is undeformed

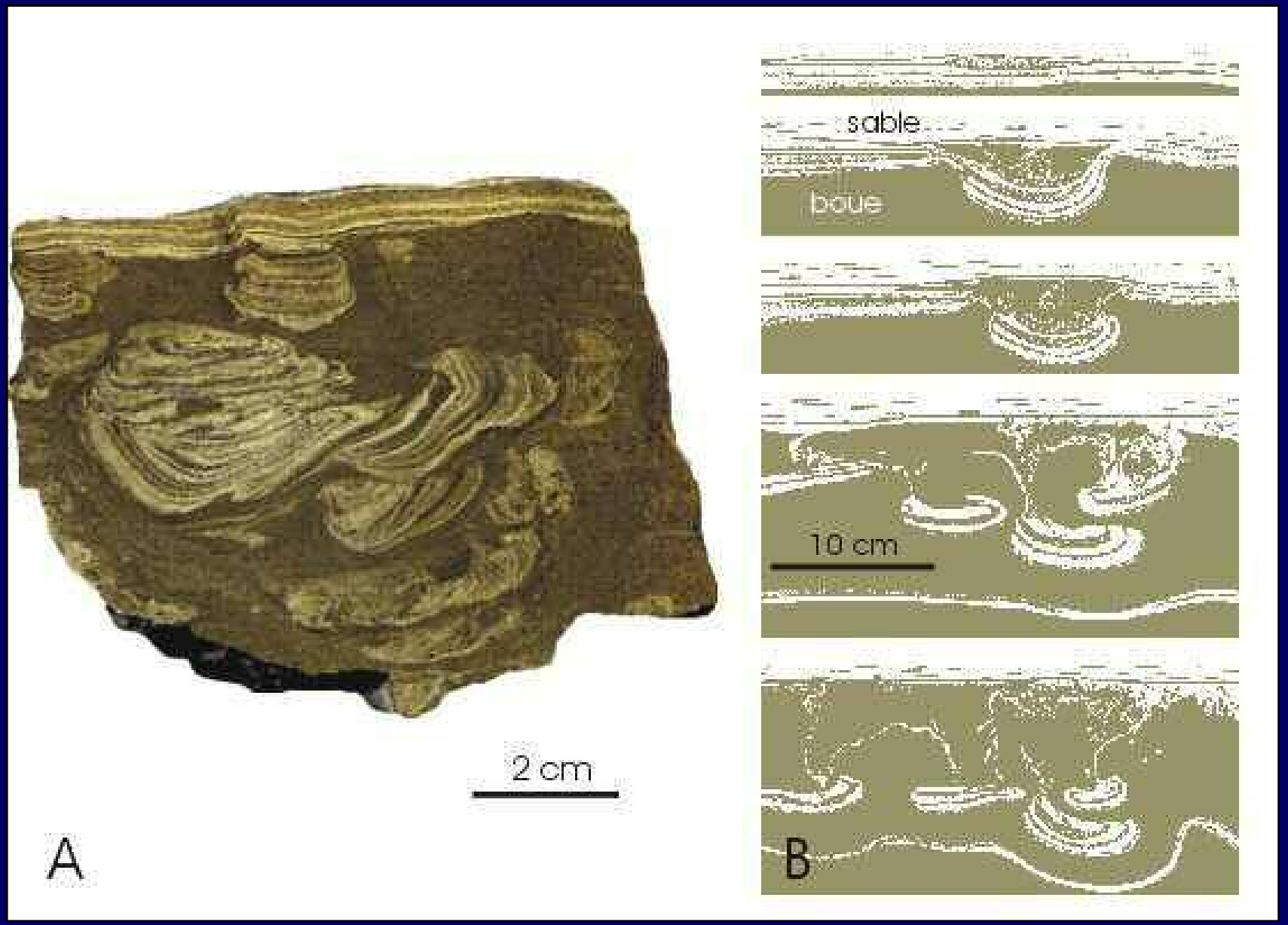
Bed B shows well developed load-casts (ball and pillow) on its lower surface

Bed C has split into discrete pseudonodules



Pseudonodules: Ball-and-pillow load structures that have become disconnected from their source bed

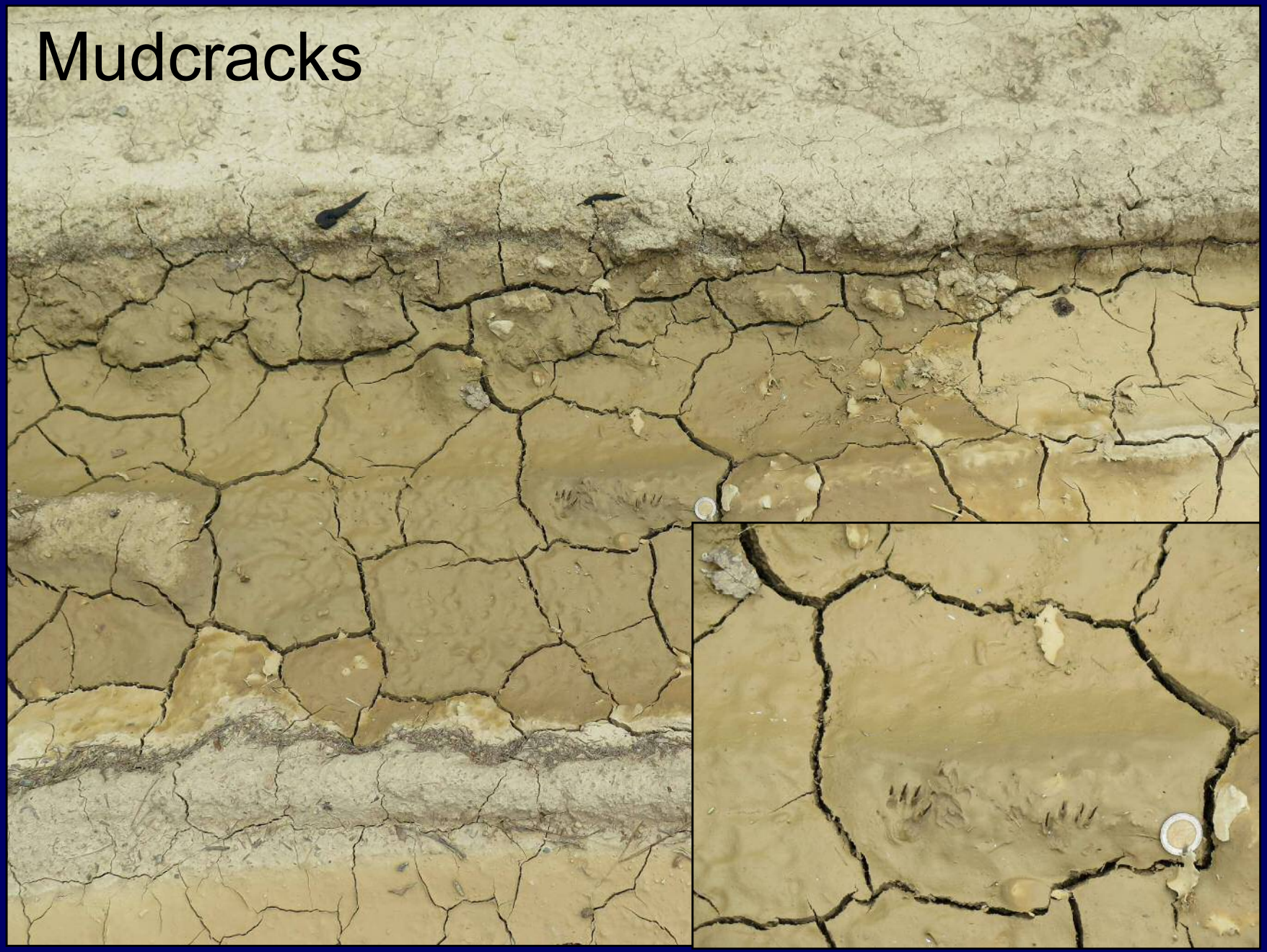
Example from floodplain mudstones, Freshwater West, Pembrokeshire



Easy to make in lab (e.g., Owen, 1992, 1996)

Maybe due to tectonic triggers (shaking)

Mudcracks

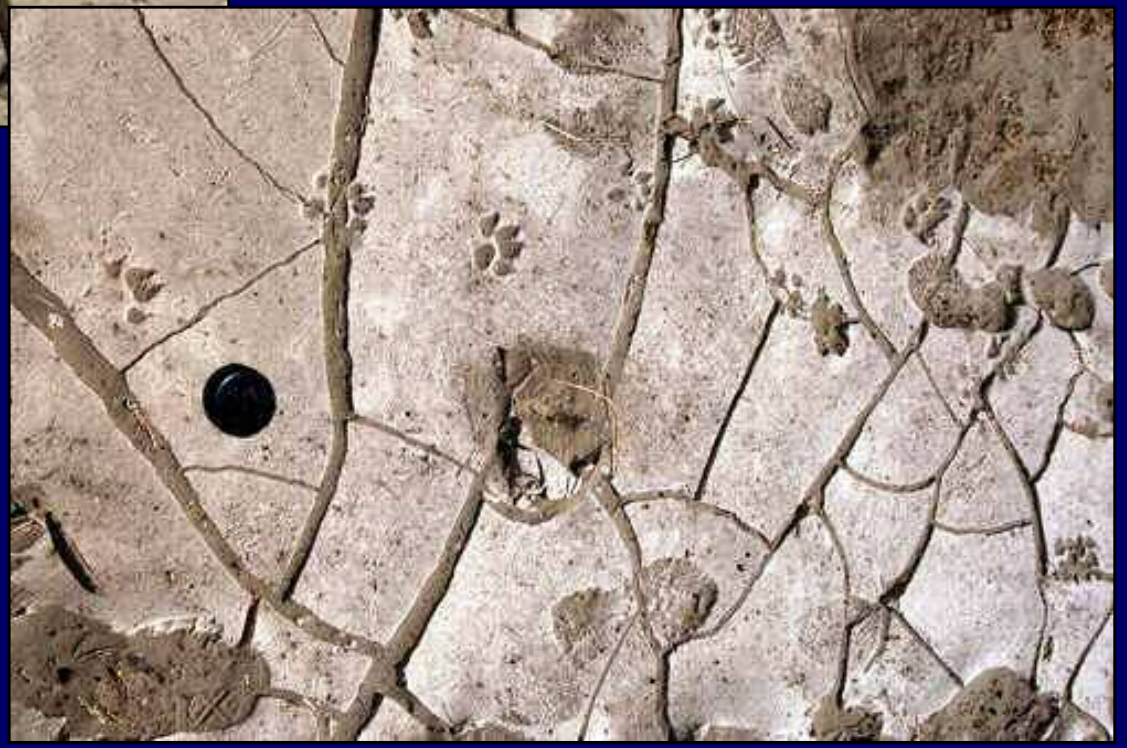


Mud shrinks when it dries out producing polygonal patterns called mudcracks or desiccation cracks



Mudcracks: why do we care?

Presence implies subaerial exposure of the sediment surface (i.e. emergence of an otherwise wet surface) NOT aridity

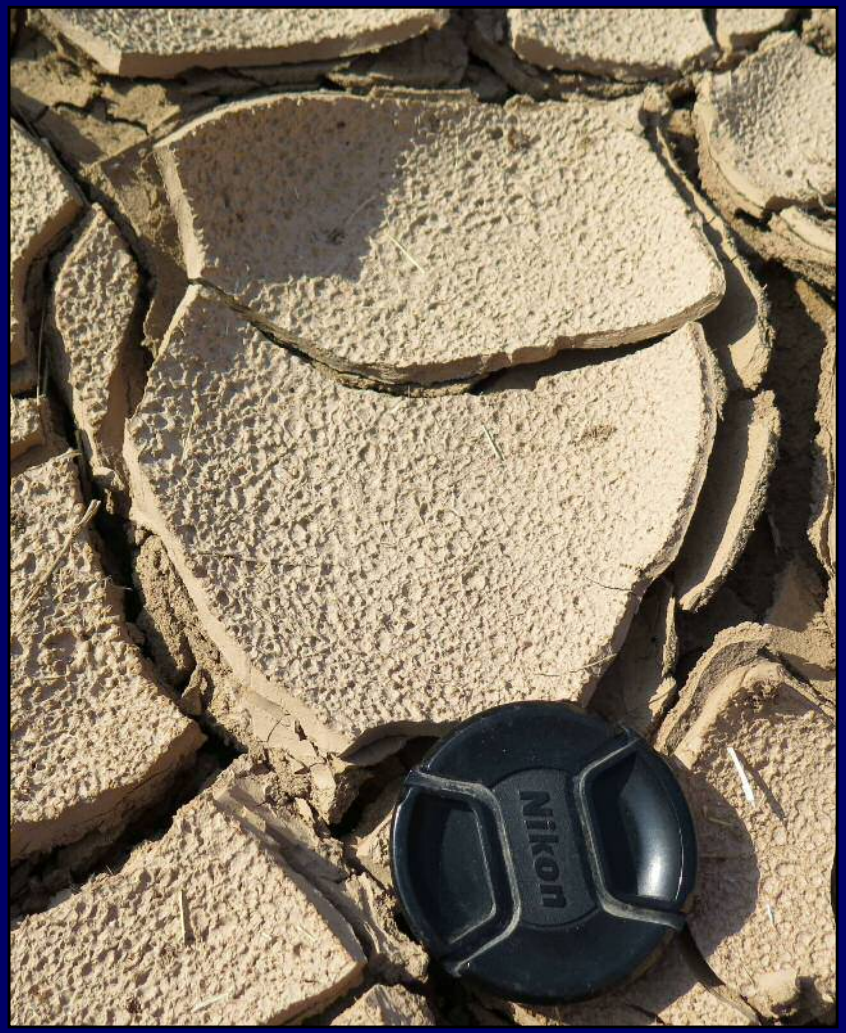


What other evidence for subaerial exposure is shown in these photos?

What type of structure are they?

Mudcracks: why do we care?





Used to determine ‘way-up’ of strata

Chips curl upwards

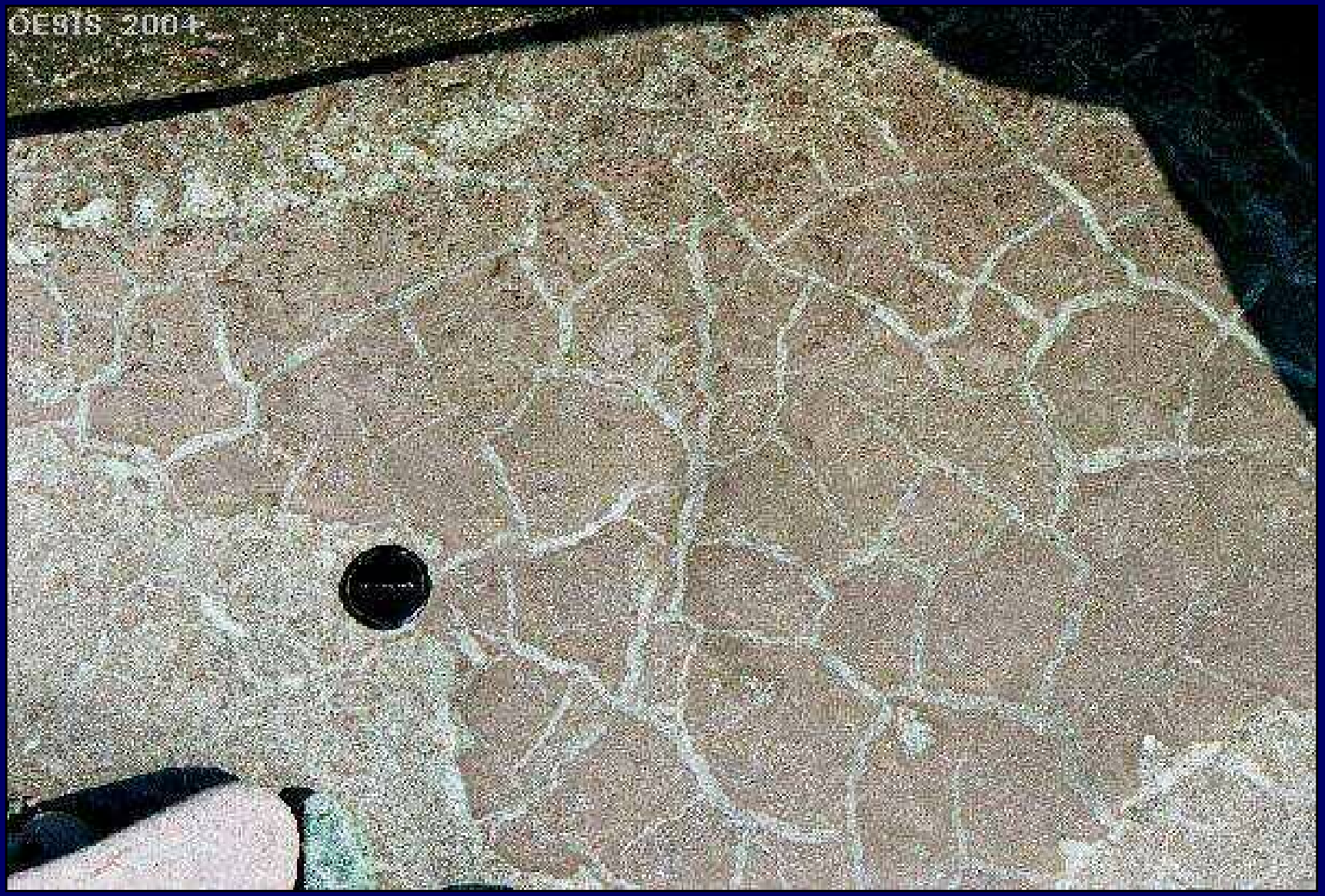
Mudcracks: why do we care?





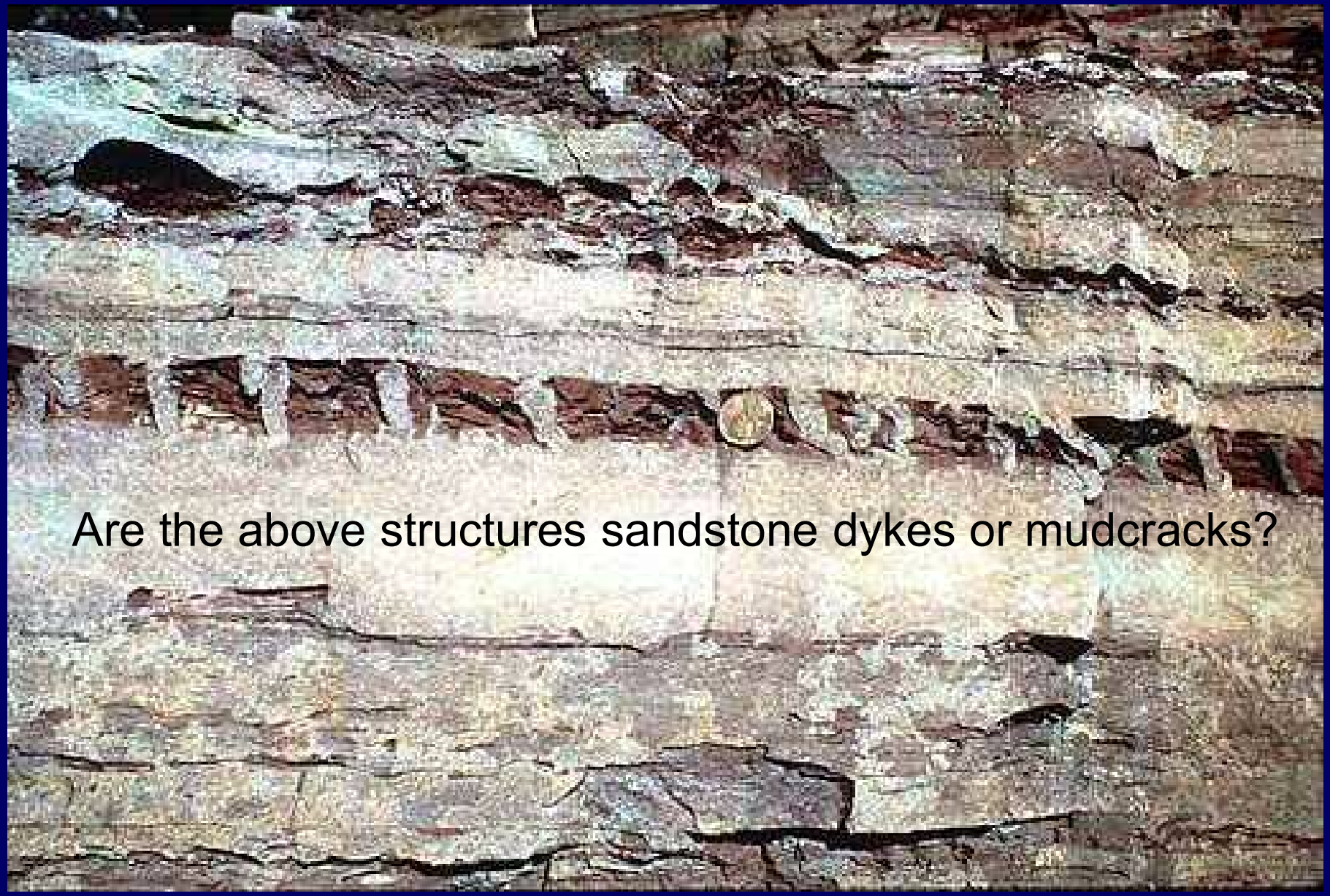
Right: Redeposited rip-up clasts on base of bed

Identification of way-up in plan view



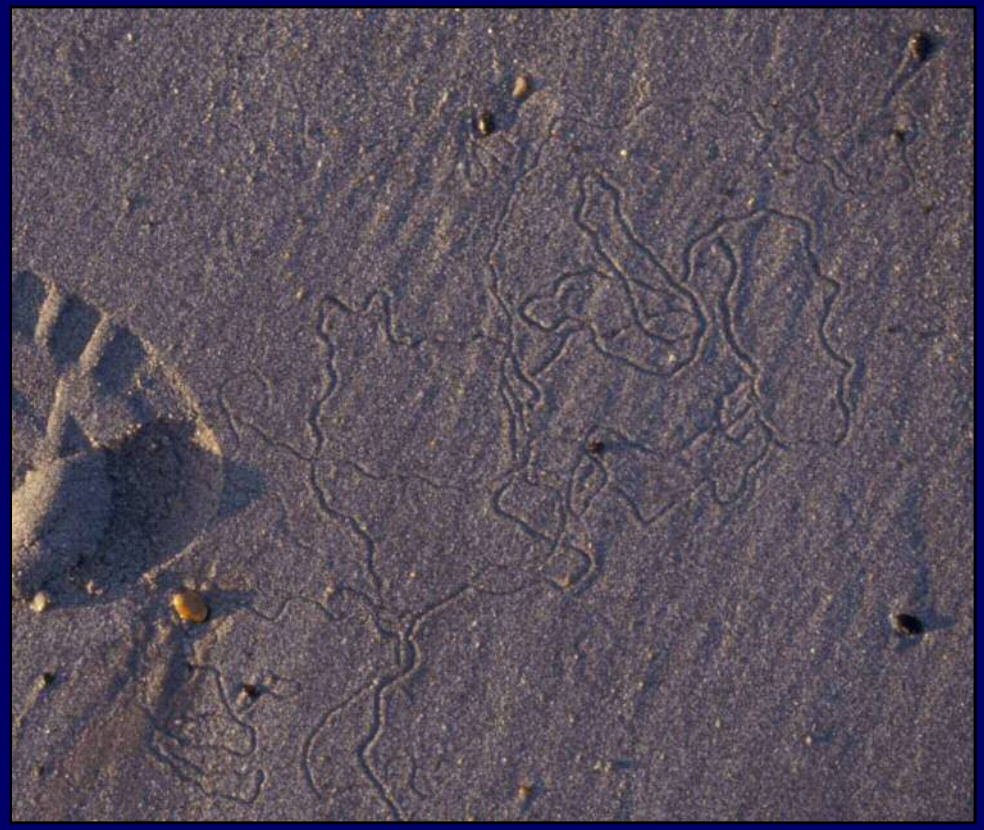
Polygonal patterns on bedding surfaces of ancient sedimentary rocks are usually infilled mudcracks

Identification of way-up in cross-section



Cracks split from the surface downwards and therefore the crack will be wider at the top and taper toward the bottom.

Biogenic structures (primary)

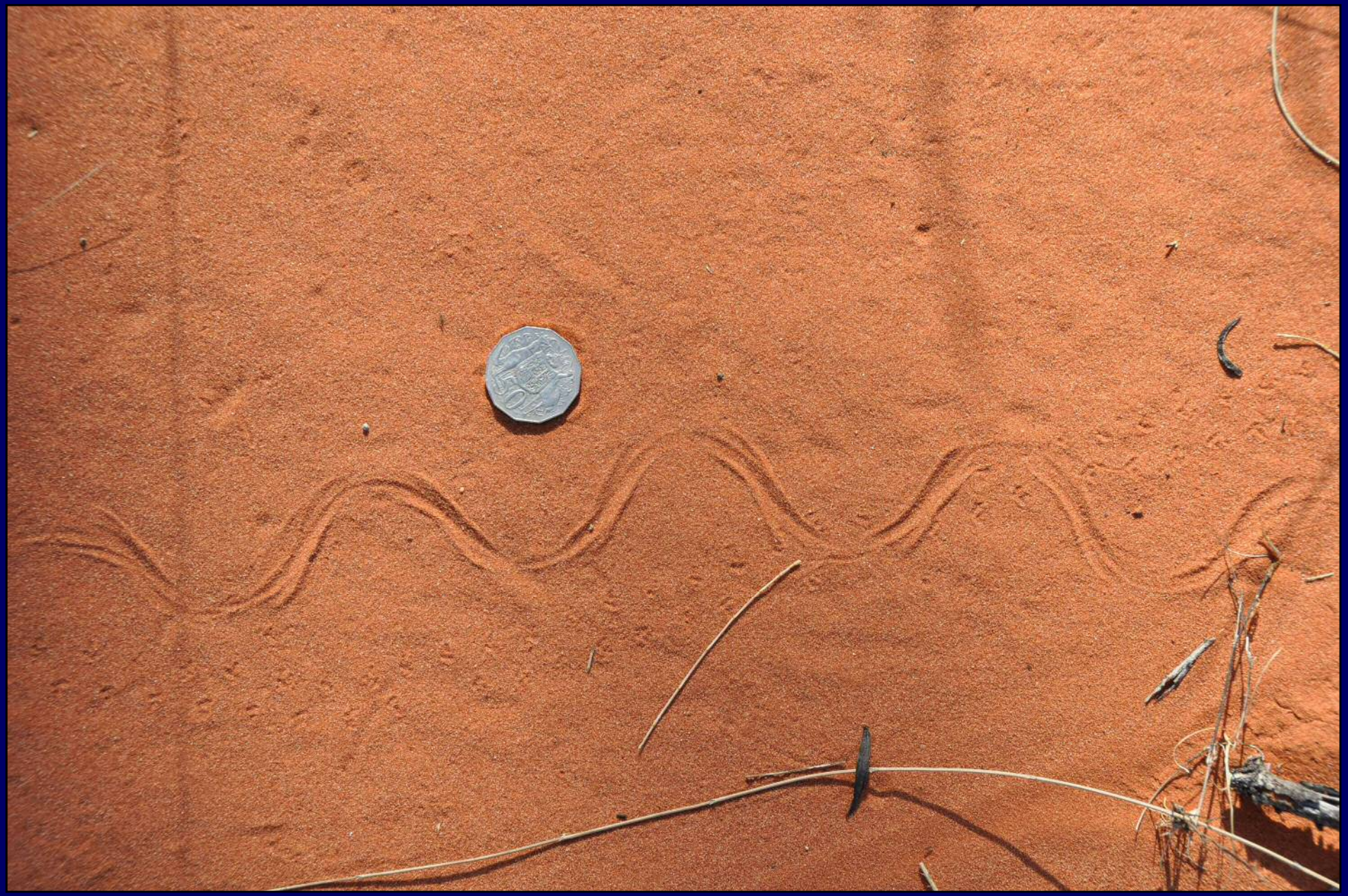






Grazing and locomotion trails:
modern and ancient

Biogenic structures: modern trackways



Biogenic structures: modern trackways



Biogenic structures: ancient trackways



Biogenic structures (secondary)



Biogenic structures: burrows



Skolithos type



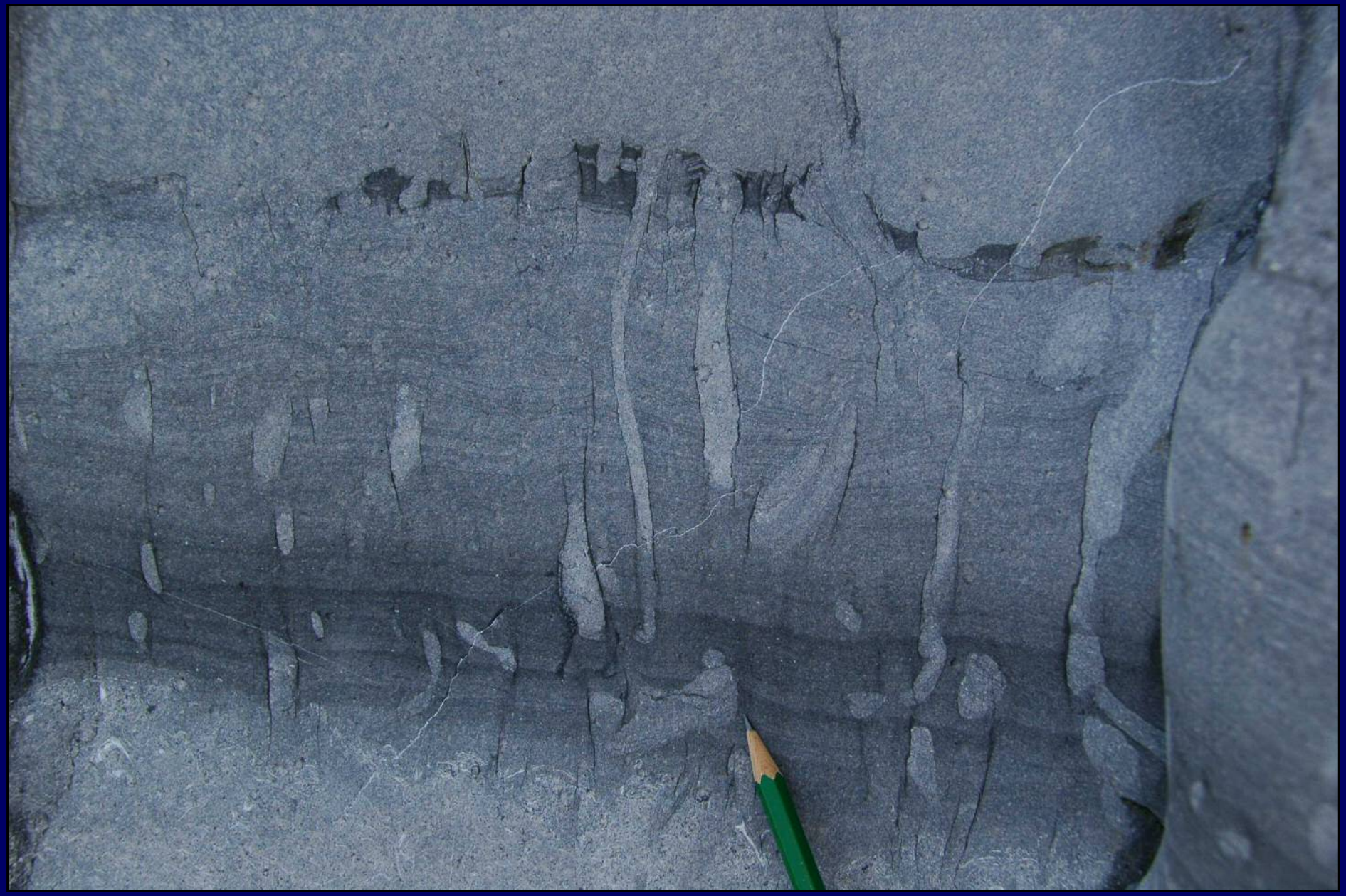
Ophiomorpha

Biogenic structures: burrows





Biogenic structures: effect of bioturbation on layering



Biogenic structures: boring



Summary

What sedimentary structures, grain size, texture and composition can tells us about sandstones and conglomerates:

- way up of bed ('younging')
- direction of flow ('palaeotransport direction')
- strength of flow and flow depth
- duration and variability of flow
- climatic conditions
- source rocks ('provenance')

Use all this information to determine the depositional environment

Suggested Reading

Nicholls, 2009, 2nd edition, chapter 4

Or if you want an alternative you could try
Tucker, 2001, 3rd edition, p.21-37
or
Boggs, 2006, 4th edition, chapter 4

For a more in depth treatment of this subject:
Allen, J.R.L., 1984, Sedimentary Structures, Their Character and Physical Basis. Elsevier.
Potter and Pettijohn, 1964, Atlas and Glossary of Primary Sedimentary Structures
Collinson, Mountney and Thompson, 2006, Sedimentary Structures, 3rd edition.
Selley, 2000, Applied Sedimentology, 2nd edition, Elsevier

Also useful for in the field
Tucker, 2011, Sedimentary rocks in the field: a practical guide (earlier editions just as good e.g., 2003)
Stow, 2005, Sedimentary rocks in the field: a colour guide
Coe et al, 2010, Geological Field Techniques